



US012387979B2

(12) **United States Patent**
Yang et al.

(10) **Patent No.:** **US 12,387,979 B2**

(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **TUNGSTEN FEATURE FILL WITH
NUCLEATION INHIBITION**

(71) Applicant: **Novellus Systems, Inc.**, Fremont, CA
(US)

(72) Inventors: **Tsung-Han Yang**, San Jose, CA (US);
Anand Chandrashekar, Fremont, CA
(US); **Jasmine Lin**, San Jose, CA (US)

(73) Assignee: **Novellus Systems, Inc.**, Fremont, CA
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **17/814,297**

(22) Filed: **Jul. 22, 2022**

(65) **Prior Publication Data**

US 2022/0359280 A1 Nov. 10, 2022

Related U.S. Application Data

(63) Continuation of application No. 16/457,353, filed on
Jun. 28, 2019, now Pat. No. 11,437,269, which is a
(Continued)

(51) **Int. Cl.**
H01L 21/768 (2006.01)
C23C 16/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H01L 21/76876** (2013.01); **C23C 16/04**
(2013.01); **C23C 16/045** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC H01L 21/043; H01L 21/0485; H01L
29/45-458; H01L 51/105;

(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,714,520 A 12/1987 Gwozdz
5,112,439 A 5/1992 Reisman et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 1798867 A 7/2006
CN 1883037 A 12/2006

(Continued)

OTHER PUBLICATIONS

Advisory Action dated Sep. 1, 2021, in U.S. Appl. No. 16/294,736.

(Continued)

Primary Examiner — Julio J Maldonado

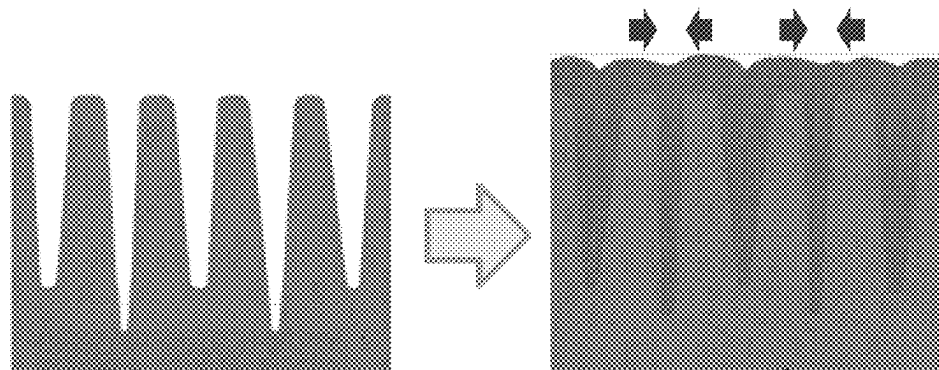
Assistant Examiner — Stanetta D Isaac

(74) *Attorney, Agent, or Firm* — Weaver Austin
Villeneuve & Sampson LLP

(57) **ABSTRACT**

Described herein are methods of filling features with tungsten, and related systems and apparatus, involving inhibition of tungsten nucleation. In some embodiments, the methods involve selective inhibition along a feature profile. Methods of selectively inhibiting tungsten nucleation can include exposing the feature to ammonia vapor in a non-plasma process. Process parameters including exposure time, substrate temperature, and chamber pressure can be used to tune the inhibition profile. Also provided are methods of filling multiple adjacent lines with reduced or no line bending. The methods involve selectively inhibiting the tungsten nucleation to reduce sidewall growth during feature fill.

19 Claims, 22 Drawing Sheets



Narrow asymmetric
trench structure typical
of DRAM BWL

Severe line bending
after W deposition

Related U.S. Application Data

continuation of application No. 15/640,940, filed on Jul. 3, 2017, now Pat. No. 10,381,266, which is a continuation-in-part of application No. 13/774,350, filed on Feb. 22, 2013, now Pat. No. 10,256,142.

- (60) Provisional application No. 62/357,526, filed on Jul. 1, 2016, provisional application No. 61/737,419, filed on Dec. 14, 2012, provisional application No. 61/616,377, filed on Mar. 27, 2012.

(51) **Int. Cl.**

C23C 16/06 (2006.01)

H01L 21/285 (2006.01)

H01L 21/321 (2006.01)

H10B 12/00 (2023.01)

(52) **U.S. Cl.**

CPC **C23C 16/06** (2013.01); **H01L 21/28556** (2013.01); **H01L 21/321** (2013.01); **H01L 21/76879** (2013.01); **H10B 12/488** (2023.02); **H01L 21/28562** (2013.01); **H01L 21/76898** (2013.01)

(58) **Field of Classification Search**

CPC H01L 21/76871-76876; H01L 2221/1084-1089; H01L 28/56; H01L 21/76841-76876; H01L 23/53223; H01L 23/53238; H01L 23/53266; H01L 29/40-518; H01L 2221/1015-1036; H01L 21/76802-76817; H01L 21/76877-76883; H01L 28/90-92; H01L 29/66181; H01L 29/945; H01L 29/7813; H01L 29/7825; H01L 29/66613-66628; H01L 29/42336; H01L 29/42352; H01L 29/4236; H01L 23/53257-53266; H01L 21/02274; H01L 21/3065-30655; H01L 21/31116; H01L 21/31122; H01L 21/31138; H01L 21/67069; H01L 21/32136-32137; H01L 21/02071; H01L 21/78; H01L 21/02252; H01L 21/02315; H01L 21/0234; H01L 21/76855; H01J 2237/334-3348; C23C 16/04; C23C 16/045; C23C 16/06; C23C 16/505; H10B 12/488

See application file for complete search history.

(56)

References Cited**U.S. PATENT DOCUMENTS**

- | | | | | | |
|--------------|---------|----------------------|-------------------|---------|---|
| 5,227,329 A | 7/1993 | Kobayashi et al. | 9,653,353 B2 | 5/2017 | Chandrashekar et al. |
| 5,407,698 A | 4/1995 | Emesh | 9,748,137 B2 | 8/2017 | Lai et al. |
| 5,504,038 A | 4/1996 | Chien et al. | 9,837,312 B1 | 12/2017 | Tan et al. |
| 5,785,796 A | 7/1998 | Lee | 9,997,405 B2 | 6/2018 | Chandrashekar et al. |
| 5,963,827 A | 10/1999 | Enomoto et al. | 10,103,058 B2 | 10/2018 | Chandrashekar et al. |
| 6,013,575 A | 1/2000 | Itoh | 10,170,320 B2 | 1/2019 | Wang et al. |
| 6,025,243 A | 2/2000 | Ohmi et al. | 10,193,059 B2 | 1/2019 | Lee et al. |
| 6,245,654 B1 | 6/2001 | Shih et al. | 10,211,099 B2 | 2/2019 | Wang et al. |
| 6,339,023 B1 | 1/2002 | Kitazaki et al. | 10,242,879 B2 | 3/2019 | Na et al. |
| 6,503,843 B1 | 1/2003 | Xia et al. | 10,256,142 B2 | 4/2019 | Chandrashekar et al. |
| 6,641,867 B1 | 11/2003 | Hsu et al. | 10,381,266 B2 | 8/2019 | Yang et al. |
| 6,790,773 B1 | 9/2004 | Drewery et al. | 10,573,522 B2 | 2/2020 | Jandl et al. |
| 7,205,240 B2 | 4/2007 | Karim et al. | 10,580,654 B2 | 3/2020 | Wang et al. |
| 7,569,913 B2 | 8/2009 | Enicks | 10,580,695 B2 | 3/2020 | Chandrashekar et al. |
| 7,655,567 B1 | 2/2010 | Gao et al. | 10,916,434 B2 | 2/2021 | Wang et al. |
| 8,124,531 B2 | 2/2012 | Chandrashekar et al. | 11,075,115 B2 | 7/2021 | Chandrashekar et al. |
| 8,129,270 B1 | 3/2012 | Chandrashekar et al. | 11,355,345 B2 | 6/2022 | Jandl et al. |
| 8,409,985 B2 | 4/2013 | Chan et al. | 11,410,883 B2 | 8/2022 | Chandrashekar et al. |
| 9,034,768 B2 | 5/2015 | Chandrashekar et al. | 11,437,269 B2 * | 9/2022 | Yang C23C 16/045 |
| 9,165,824 B2 | 10/2015 | Chandhok et al. | 11,901,227 B2 | 2/2024 | Chandrashekar et al. |
| 9,240,347 B2 | 1/2016 | Chandrashekar et al. | 11,978,666 B2 | 5/2024 | Chandrashekar et al. |
| 9,349,637 B2 | 5/2016 | Na et al. | 12,261,081 B2 | 3/2025 | Yang et al. |
| | | | 2002/0090811 A1 | 7/2002 | Kim et al. |
| | | | 2002/0132472 A1 * | 9/2002 | Park H01L 21/76879
257/E21.586 |
| | | | 2002/0192953 A1 | 12/2002 | Wang et al. |
| | | | 2003/0042465 A1 | 3/2003 | Ko |
| | | | 2003/0051665 A1 | 3/2003 | Zhao et al. |
| | | | 2003/0082902 A1 | 5/2003 | Fukui et al. |
| | | | 2003/0092280 A1 | 5/2003 | Lee et al. |
| | | | 2003/0190802 A1 | 10/2003 | Wang et al. |
| | | | 2003/0194850 A1 | 10/2003 | Lewis et al. |
| | | | 2003/0203123 A1 | 10/2003 | Shang et al. |
| | | | 2003/0222346 A1 | 12/2003 | Yun et al. |
| | | | 2004/0120095 A1 | 6/2004 | Yanagida |
| | | | 2004/0149386 A1 | 8/2004 | Numasawa et al. |
| | | | 2005/0014365 A1 | 1/2005 | Moon et al. |
| | | | 2005/0085070 A1 * | 4/2005 | Park H01L 21/76856
257/E21.586 |
| | | | 2005/0147762 A1 | 7/2005 | Dubin et al. |
| | | | 2005/0179141 A1 | 8/2005 | Yun et al. |
| | | | 2006/0009034 A1 | 1/2006 | Lai et al. |
| | | | 2006/0075966 A1 | 4/2006 | Chen et al. |
| | | | 2006/0265100 A1 | 11/2006 | Li |
| | | | 2007/0066060 A1 | 3/2007 | Wang |
| | | | 2007/0166989 A1 | 7/2007 | Fresco et al. |
| | | | 2007/0199922 A1 | 8/2007 | Shen et al. |
| | | | 2007/0264105 A1 | 11/2007 | Pharand et al. |
| | | | 2008/0054468 A1 | 3/2008 | Choi et al. |
| | | | 2008/0079156 A1 | 4/2008 | Choi et al. |
| | | | 2008/0081470 A1 | 4/2008 | Clark |
| | | | 2008/0254623 A1 | 10/2008 | Chan et al. |
| | | | 2008/0268635 A1 | 10/2008 | Yu et al. |
| | | | 2009/0004850 A1 | 1/2009 | Ganguli et al. |
| | | | 2009/0053426 A1 | 2/2009 | Lu et al. |
| | | | 2009/0053893 A1 | 2/2009 | Khandelwal et al. |
| | | | 2009/0149022 A1 | 6/2009 | Chan et al. |
| | | | 2009/0163025 A1 | 6/2009 | Humayun et al. |
| | | | 2009/0183984 A1 | 7/2009 | Sakuma et al. |
| | | | 2010/0072623 A1 | 3/2010 | Prindle et al. |
| | | | 2010/0130002 A1 | 5/2010 | Dao et al. |
| | | | 2010/0144140 A1 | 6/2010 | Chandrashekar et al. |
| | | | 2010/0159694 A1 | 6/2010 | Chandrashekar et al. |
| | | | 2010/0167527 A1 | 7/2010 | Wu et al. |
| | | | 2010/0267230 A1 | 10/2010 | Chandrashekar et al. |
| | | | 2010/0267235 A1 | 10/2010 | Chen et al. |
| | | | 2011/0045672 A1 | 2/2011 | Srinivasan et al. |
| | | | 2011/0111533 A1 | 5/2011 | Varadarajan et al. |
| | | | 2011/0151670 A1 | 6/2011 | Lee et al. |
| | | | 2011/0159690 A1 | 6/2011 | Chandrashekar et al. |
| | | | 2011/0223763 A1 | 9/2011 | Chan et al. |
| | | | 2011/0233778 A1 | 9/2011 | Lee et al. |
| | | | 2011/0256717 A1 | 10/2011 | Choi et al. |
| | | | 2011/0256726 A1 | 10/2011 | LaVoie et al. |
| | | | 2012/0009785 A1 | 1/2012 | Chandrashekar et al. |
| | | | 2012/0077342 A1 | 3/2012 | Gao et al. |
| | | | 2012/0149213 A1 | 6/2012 | Nittala et al. |
| | | | 2012/0177845 A1 | 7/2012 | Odedra et al. |
| | | | 2012/0187305 A1 | 7/2012 | Elam et al. |

(56)	References Cited			CN	104081502	A	10/2014
	U.S. PATENT DOCUMENTS			CN	104272440	A	1/2015
				CN	104513973	A	4/2015
				CN	104752339	A	7/2015
2013/0062677	A1	3/2013	Li et al.	CN	104975268	A	10/2015
2013/0171822	A1	7/2013	Chandrashekar et al.	CN	105390438	A	3/2016
2013/0260555	A1	10/2013	Zope et al.	CN	105405764	A	3/2016
2014/0087587	A1	3/2014	Lind	CN	106169440	A	11/2016
2014/0106083	A1	4/2014	Wu et al.	CN	107768304	A	3/2018
2014/0120723	A1	5/2014	Fu et al.	CN	110004429	B	8/2021
2014/0231896	A1	8/2014	Matsumori et al.	EP	0430303	A2	6/1991
2014/0264932	A1	9/2014	Ting et al.	EP	1672687	A1	6/2006
2015/0091175	A1	4/2015	Chandhok et al.	EP	1961838	A1	8/2008
2015/0093890	A1	4/2015	Blackwell et al.	EP	2779224	A2	9/2014
2015/0162214	A1	6/2015	Thompson et al.	JP	H0225568	A	1/1990
2015/0294906	A1	10/2015	Wu et al.	JP	H02187031	A	7/1990
2015/0361547	A1	12/2015	Lin et al.	JP	H03110840	A	5/1991
2016/0017493	A1	1/2016	Dhas et al.	JP	H04142061	A	5/1992
2016/0020169	A1	1/2016	Matsuda	JP	H05226280	A	9/1993
2016/0056074	A1	2/2016	Na et al.	JP	H0794488	A	4/1995
2016/0056077	A1	2/2016	Lai et al.	JP	H07147321	A	6/1995
2016/0093528	A1	3/2016	Chandrashekar et al.	JP	H07226393	A	8/1995
2016/0118345	A1	4/2016	Chen et al.	JP	H0922896	A	1/1997
2016/0172211	A1	6/2016	Demos et al.	JP	2002016066	A	1/2002
2016/0177443	A1	6/2016	Kumar et al.	JP	2002294449	A	10/2002
2016/0190008	A1	6/2016	Chandrashekar et al.	JP	2003022985	A	1/2003
2016/0293467	A1	10/2016	Caveney et al.	JP	2005129831	A	5/2005
2016/0343612	A1	11/2016	Wang et al.	JP	2007184329	A	7/2007
2017/0053811	A1	2/2017	Fung et al.	JP	2007251164	A	9/2007
2017/0125548	A1	5/2017	Hung et al.	JP	2008049976	A	3/2008
2017/0229341	A1	8/2017	Chang et al.	JP	2008177577	A	7/2008
2017/0278749	A1	9/2017	Chandrashekar et al.	JP	2009024252	A	2/2009
2017/0365513	A1	12/2017	Yang et al.	JP	2010251760	A	11/2010
2018/0053660	A1	2/2018	Jandl et al.	JP	2010541252	A	12/2010
2018/0142345	A1	5/2018	Meng et al.	JP	2011035366	A	2/2011
2018/0174901	A1	6/2018	Wang et al.	JP	2011199021	A	10/2011
2018/0277431	A1	9/2018	Chandrashekar et al.	JP	2011228709	A	11/2011
2018/0308701	A1	10/2018	Na et al.	JP	2014063926	A	4/2014
2018/0315650	A1	11/2018	Ren et al.	JP	2014183185	A	9/2014
2019/0019725	A1	1/2019	Chandrashekar et al.	JP	2015514160	A	5/2015
2019/0080914	A1	3/2019	Wang et al.	JP	2016025141	A	2/2016
2019/0206731	A1	7/2019	Chandrashekar et al.	JP	2016082069	A	5/2016
2019/0326168	A1	10/2019	Yang et al.	JP	2018006753	A	1/2018
2020/0083251	A1	3/2020	Lee	KR	19980011846	U	5/1998
2020/0144066	A1	5/2020	Jandl et al.	KR	20010030488	A	4/2001
2020/0185225	A1	6/2020	Wang et al.	KR	20010048302	A	6/2001
2020/0185273	A1	6/2020	Chandrashekar et al.	KR	20030092583	A	12/2003
2021/0028187	A1	1/2021	Byeon et al.	KR	20050008320	A	1/2005
2021/0125832	A1	4/2021	Bhatnagar	KR	20050022261	A	3/2005
2021/0327754	A1	10/2021	Chandrashekar et al.	KR	20050087428	A	8/2005
2022/0020641	A1	1/2022	Chandrashekar et al.	KR	20060087844	A	8/2006
2022/0102208	A1	3/2022	Chandrashekar et al.	KR	20070054100	A	5/2007
2022/0172987	A1	6/2022	Yang et al.	KR	100757418	B1	9/2007
2023/0041794	A1	2/2023	Chandrashekar et al.	KR	20080001460	A	1/2008
2023/0122846	A1	4/2023	Khare et al.	KR	20080061978	A	7/2008
2023/0130557	A1	4/2023	Birru et al.	KR	20080094088	A	10/2008
2024/0234208	A1	7/2024	Chandrashekar et al.	KR	20090068187	A	6/2009
2025/0038050	A1	1/2025	Tran et al.	KR	20090074560	A	7/2009
				KR	20100067065	A	6/2010
				KR	20100114856	A	10/2010
				KR	20110052436	A	5/2011
				KR	20110108382	A	10/2011
CN	200956667	Y	10/2007	KR	20120005992	A	1/2012
CN	101154576	A	4/2008	KR	20120043077	A	5/2012
CN	101213320	A	7/2008	KR	20140141686	A	12/2014
CN	101308794	A	11/2008	KR	20140143202	A	12/2014
CN	101313085	A	11/2008	KR	20150063562	A	6/2015
CN	101447427	A	6/2009	KR	20160039139	A	4/2016
CN	101764044	A	6/2010	KR	20160044004	A	4/2016
CN	101770978	A	7/2010	KR	20160072054	A	6/2016
CN	101789369	A	7/2010	KR	20160108174	A	9/2016
CN	101899649	A	12/2010	TW	201120233	A	6/2011
CN	102007573	A	4/2011	TW	201130045	A	9/2011
CN	102074500	A	5/2011	TW	201405707	A	2/2014
CN	102224574	A	10/2011	TW	201405781	A	2/2014
CN	102574884	A	7/2012	TW	201413031	A	4/2014
CN	102892922	A	1/2013	TW	201525173	A	7/2015
CN	103069550	A	4/2013	TW	201626503	A	7/2016
CN	103107120	A	5/2013	WO	WO-2007027350	A2	3/2007
CN	103125013	A	5/2013				

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	WO-2009042913	A1	4/2009
WO	WO-2011027834	A1	3/2011
WO	WO-2013148444	A1	10/2013
WO	WO-2013148490	A1	10/2013
WO	WO-2013148880	A1	10/2013
WO	WO-2014105477	A1	7/2014
WO	WO-2020168071	A1	8/2020

OTHER PUBLICATIONS

Advisory Action dated Sep. 1, 2021, in U.S. Appl. No. 16/457,353.
 Chinese Fifth Office Action dated Aug. 24, 2018 issued in Application No. CN 201380022693.8.
 Chinese First Office Action dated Apr. 2, 2021 issued in Application No. CN 201910499775.0.
 Chinese First Office Action dated Dec. 12, 2017 issued in Application No. CN 201510518752.1.
 Chinese First Office Action dated Dec. 20, 2017 issued in Application No. CN 201510644832.1.
 Chinese First Office Action dated Feb. 26, 2016, issued in CN 201380022648.2.
 Chinese First Office Action dated Jan. 4, 2021 issued in Application No. CN 201710700258.6.
 Chinese first Office Action dated Jul. 27, 2018 issued in Application No. CN 201610332922.1.
 Chinese First Office Action dated Jun. 3, 2019 issued in Application No. CN 201711372325.2.
 Chinese First Office Action dated Mar. 18, 2016 issued in Application No. CN 201380022693.8.
 Chinese First Office Action dated Oct. 27, 2020 issued in Application No. CN 201811491805.5.
 Chinese Fourth Office Action dated Mar. 15, 2018 issued in Application No. CN 201380022693.8.
 Chinese Second Office Action dated Aug. 17, 2018 issued in Application No. CN 201510644832.1.
 Chinese Second Office Action dated Aug. 24, 2018 issued in Application No. CN 201510518752.1.
 Chinese Second Office Action dated Jan. 23, 2017 issued in Application No. CN 201380022693.8.
 Chinese Second Office Action dated Jun. 17, 2021 issued in Application No. CN 201710700258.6.
 Chinese Second Office Action dated May 7, 2020 issued in Application No. CN 201711372325.2.
 Chinese Third Office Action dated Dec. 1, 2020 issued in Application No. CN 201711372325.2.
 Chinese Third Office Action dated Feb. 14, 2019 issued in Application No. CN 201510644832.1.
 Chinese Third Office Action dated Sep. 25, 2017 issued in Application No. CN 201380022693.8.
 Chinese Third Office [Decision of Final Rejection] Action dated Mar. 6, 2019 issued in Application No. CN 201510518752.1.
 CN Office action dated Apr. 7, 2022 in CN Application No. CN201910499775.
 CN Office action dated Oct. 25, 2021 in CN Application No. CN201910499775 with English translation.
 Co-pending U.S. Appl. No. 17/497,702, filed Oct. 8, 2021.
 International Preliminary Report on Patentability dated Jun. 17, 2021 issued in Application No. PCT/US2019/064768.
 International Search Report and Written Opinion dated Mar. 27, 2020 issued in Application No. PCT/US2019/064768.
 Japanese First Office Action dated Jun. 13, 2019 issued in Application No. JP 2015-162354.
 Japanese Notification of Reasons for Rejection dated Dec. 20, 2016 issued in Application No. JP 2015-503547.
 Japanese Office Action dated Jan. 24, 2017, issued in JP 2015-514160.
 Japanese Second Office Action dated Feb. 16, 2020 issued in Application No. JP 2015-162354.

Korean Decision for Grant dated Apr. 7, 2020 issued in Application No. KR 10-2020-7000199.
 Korean Decision for Grant dated Jan. 15, 2020 issued in Application No. KR 10-2014-7029798.
 Korean Final Office Action dated Sep. 19, 2017, issued in Application No. KR 10-2013-0054726.
 Korean First Office Action dated Apr. 18, 2019 issued in Application No. KR 10-2014-7030125.
 Korean First Office Action dated Sep. 26, 2016, issued in Application No. KR 10-2013-0054726.
 Korean Office Action dated Jun. 18, 2019 issued in Application No. KR 10-2014-7029798.
 Korean Provisional Rejection dated Nov. 16, 2012, issued in Application No. KR 2011-0068603.
 Korean Second Office Action dated May 30, 2017, issued in Application No. KR 10-2013-0054726.
 Korean Second Office Action dated Oct. 25, 2019 issued in Application No. KR 10-2014-7029798.
 KR Office Action dated Aug. 1, 2022, in Application No. 10-2017-0172906.
 KR Office Action dated Feb. 8, 2022, in Application No. 10-2017-0172906 with English translation.
 KR Office Action dated Feb. 21, 2022, in Application No. KR10-2017-0102113 with English Translation.
 KR Office Action dated Jun. 26, 2022 in Application No. KR10-2015-0137906 with English translation.
 Notice of Allowance dated Oct. 5, 2020 issued in U.S. Appl. No. 16/786,513.
 Office Action dated Mar. 27, 2020 issued in U.S. Appl. No. 16/786,513.
 PCT International Preliminary Report on Patentability and Written Opinion, dated Oct. 9, 2014, issued in PCT/US2013/033174.
 PCT International Preliminary Report on Patentability and Written Opinion, dated Oct. 9, 2014, issued in PCT/US2013/034167.
 PCT International Search Report and Written Opinion, dated Jul. 26, 2013, issued in PCT/US2013/034167.
 PCT International Search Report and Written Opinion, dated Jun. 28, 2013, issued in PCT/US2013/033174.
 Taiwan Office Action and Search Report dated Nov. 22, 2016 issued in Application No. TW 102110937.
 Taiwan Office Action dated Oct. 25, 2016 issued in Application No. TW 102110947.
 Taiwanese First Office Action dated Apr. 18, 2019 issued in Application No. TW 104127083.
 Taiwanese First Office Action dated Apr. 21, 2021 issued in Application No. TW 106144306.
 Taiwanese First Office Action dated Jul. 23, 2021 issued in Application No. TW 107113100.
 Taiwanese First Office Action dated Mar. 28, 2019 issued in Application No. TW 104132010.
 Taiwanese First Office Action dated Sep. 5, 2019 issued in Application No. TW 105115121.
 Taiwanese Second Office Action dated Feb. 6, 2020 issued in Application No. TW 104132010.
 Taiwanese Second Office Action dated Mar. 16, 2020 issued in Application No. TW 104127083.
 U.S. Corrected Notice of Allowance dated Jun. 30, 2022 in U.S. Appl. No. 16/294,736.
 U.S. Corrected Notice of Allowance dated Aug. 8, 2022 in U.S. Appl. No. 16/457,353.
 U.S. Corrected Notice of Allowance dated Feb. 16, 2022, in U.S. Appl. No. 16/724,231.
 U.S. Corrected Notice of Allowance dated May 11, 2022, in U.S. Appl. No. 16/724,231.
 US Final Office Action, dated Apr. 14, 2017, issued in U.S. Appl. No. 14/965,806.
 US Final Office Action dated Apr. 8, 2021 issued in U.S. Appl. No. 16/793,464.
 US Final Office Action dated Aug. 13, 2018 issued in U.S. Appl. No. 15/492,976.
 US Final Office Action dated Dec. 15, 2020 issued in U.S. Appl. No. 16/124,050.

(56)

References Cited

OTHER PUBLICATIONS

US Final Office Action dated Feb. 6, 2020 issued in U.S. Appl. No. 16/124,050.
 US Final Office Action, dated Jan. 18, 2017, issued in U.S. Appl. No. 13/774,350.
 US Final Office Action, dated Jul. 17, 2015, issued in U.S. Appl. No. 14/502,817.
 US Final Office Action dated Jul. 5, 2019 issued in U.S. Appl. No. 15/991,413.
 US Final Office Action, dated Jul. 6, 2018, issued in U.S. Appl. No. 13/774,350.
 US Final Office Action dated Jun. 10, 2019 issued in U.S. Appl. No. 15/673,320.
 US Final Office Action, dated Jun. 14, 2021, issued in U.S. Appl. No. 16/294,736.
 US Final Office Action dated Jun. 14, 2021, issued in U.S. Appl. No. 16/457,353.
 US Final Office Action, dated Jun. 18, 2018, issued in U.S. Appl. No. 15/384,175.
 US Final Office Action dated Mar. 29, 2021 issued in U.S. Appl. No. 16/724,231.
 US Final Office Action, dated Nov. 2, 2018, issued in U.S. Appl. No. 15/640,940.
 US Final Office Action dated Oct. 19, 2017 issued in U.S. Appl. No. 14/866,621.
 US Notice of Allowance dated Apr. 27, 2017 issued in U.S. Appl. No. 14/873,152.
 U.S. Notice of Allowance dated Apr. 27, 2022 in U.S. Appl. No. 16/457,353.
 US Notice of Allowance dated Aug. 10, 2018 issued in U.S. Appl. No. 15/156,129.
 US Notice of Allowance, dated Aug. 3, 2016, issued in U.S. Appl. No. 13/851,885.
 US Notice of Allowance, dated Dec. 14, 2016, issued in U.S. Appl. No. 13/851,885.
 US Notice of Allowance dated Feb. 13, 2018 issued in U.S. Appl. No. 14/866,621.
 U.S. Notice of Allowance dated Feb. 2, 2022 in U.S. Appl. No. 16/724,231.
 US Notice of Allowance dated Jan. 14, 2016 issued in U.S. Appl. No. 14/465,610.
 US Notice of Allowance dated Jan. 9, 2012 issued in U.S. Appl. No. 13/016,656.
 US Notice of Allowance, dated Jun. 6, 2018, issued in U.S. Appl. No. 15/482,271.
 US Notice of Allowance, dated Mar. 12, 2019, issued in U.S. Appl. No. 15/640,940.
 US Notice of Allowance dated Mar. 17, 2021 issued in U.S. Appl. No. 16/124,050.
 U.S. Notice of Allowance dated Mar. 23, 2022 in U.S. Appl. No. 16/294,736.
 US Notice of Allowance, dated Mar. 24, 2017, issued in U.S. Appl. No. 13/851,885.
 US Notice of Allowance dated Nov. 13, 2018 issued in U.S. Appl. No. 15/492,976.
 US Notice of Allowance, dated Nov. 30, 2018, issued in U.S. Appl. No. 13/774,350.
 US Notice of Allowance dated Oct. 21, 2016 issued in U.S. Appl. No. 14/866,621.
 US Notice of Allowance dated Oct. 23, 2019 issued in U.S. Appl. No. 16/189,368.
 US Notice of Allowance dated Oct. 24, 2019 issued in U.S. Appl. No. 15/991,413.
 US Notice of Allowance, dated Oct. 3, 2018, issued in U.S. Appl. No. 15/384,175.
 US Notice of Allowance, dated Sep. 25, 2015, issued in U.S. Appl. No. 14/502,817.
 US Notice of Allowance dated Sep. 26, 2019 issued in U.S. Appl. No. 15/673,320.

US Notice of allowance dated Sep. 29, 2021 issued in U.S. Appl. No. 16/724,231.
 US Office Action dated Apr. 6, 2020, issued in U.S. Appl. No. 16/457,353.
 US Office Action dated Apr. 8, 2019 issued in U.S. Appl. No. 16/189,368.
 US Office Action, dated Dec. 12, 2017, issued in U.S. Appl. No. 15/482,271.
 US Office Action dated Dec. 15, 2017 issued in U.S. Appl. No. 15/156,129.
 US Office Action, dated Dec. 18, 2014, issued in U.S. Appl. No. 14/502,817.
 US Office Action dated Dec. 21, 2018 issued in U.S. Appl. No. 15/673,320.
 US Office Action, dated Dec. 23, 2014, issued in U.S. Appl. No. 13/851,885.
 US Office Action, dated Dec. 7, 2020, issued in U.S. Appl. No. 16/294,736.
 US Office Action dated Dec. 8, 2020, issued in U.S. Appl. No. 16/457,353.
 US Office Action dated Feb. 5, 2019 issued in U.S. Appl. No. 15/991,413.
 US Office Action, dated Jan. 15, 2015, issued in U.S. Appl. No. 13/774,350.
 US Office Action, dated Jun. 2, 2016, issued in U.S. Appl. No. 13/774,350.
 US Office Action dated Jun. 23, 2020 issued in U.S. Appl. No. 16/124,050.
 US Office Action dated Mar. 21, 2018 issued in U.S. Appl. No. 15/492,976.
 US Office Action, dated Mar. 26, 2018, issued in U.S. Appl. No. 15/640,940.
 US Office Action, dated May 15, 2020, issued in U.S. Appl. No. 16/294,736.
 US Office Action dated May 18, 2017 issued in U.S. Appl. No. 14/866,621.
 US Office Action dated May 25, 2016 issued in U.S. Appl. No. 14/866,621.
 US Office Action dated Nov. 1, 2016 issued in U.S. Appl. No. 14/873,152.
 US Office Action, dated Nov. 20, 2017, issued in U.S. Appl. No. 15/384,175.
 US Office Action dated Nov. 25, 2011 issued in U.S. Appl. No. 13/016,656.
 US Office Action dated Oct. 30, 2020 issued in U.S. Appl. No. 16/793,464.
 US Office Action, dated Oct. 8, 2015, issued in U.S. Appl. No. 13/774,350.
 US Office Action, dated Sep. 11, 2017, issued in U.S. Appl. No. 14/965,806.
 US Office Action, dated Sep. 2, 2016, issued in U.S. Appl. No. 14/965,806.
 US Office Action, dated Sep. 20, 2017, issued in U.S. Appl. No. 13/774,350.
 US Office Action dated Sep. 22, 2020 issued in U.S. Appl. No. 16/724,231.
 US Office Action, dated Sep. 24, 2019, issued in U.S. Appl. No. 16/124,050.
 US Office Action dated Sep. 3, 2015 issued in U.S. Appl. No. 14/465,610.
 U.S. Appl. No. 17/809,480, inventors Chandrashekar et al., filed Jun. 28, 2022.
 Van Zant, P., "Microchip fabrication: a practical guide to semiconductor processing" 4th ed., (2000) p. 263. [ISBN 0-07-135636-3].
 Williams et al., "Etch Rates for Micromachining Processing-Part II" Journal of Microelectromechanical Systems, vol. 12, No. 6, Dec. 2003, pp. 761-778.
 CN Office action dated Aug. 2, 2022 in CN Application No. CN201910499775 with English translation.
 CN Office action dated Apr. 7, 2022 in CN Application No. CN201910499775 with English translation.
 CN Office Action dated Aug. 16, 2023, in Application No. CN201810358633.8 with English translation.

(56)

References Cited**OTHER PUBLICATIONS**

- CN Office Action dated Dec. 8, 2022, in Application No. CN201980081000.X with English translation.
- CN Office Action dated Feb. 15, 2023, in Application No. CN201810358633.8 with English translation.
- CN Office Action dated Jun. 29, 2023, in Application No. CN201980081000.X with English translation.
- CN Office Action dated Mar. 31, 2023, in Application No. CN201910729470.4 with English translation.
- CN Office Action dated Oct. 26, 2022 in Application No. CN201910729470 With English Translation.
- International Preliminary Report on Patentability and written opinion dated Sep. 15, 2022, in PCT Application No. PCT/US2021/020748.
- International Search Report and Written Opinion dated Jul. 1, 2021 in Application No. PCT/US2021/022152.
- International Preliminary Report on Patentability dated Oct. 6, 2022, in PCT Application No. PCT/US2021/022152.
- International Search Report and Written Opinion dated Jun. 18, 2021, in PCT Application No. PCT/US2021/020748.
- Kim, C. et al., "Pulsed CVD-W Nucleation Layer Using WF₆ and B₂H₆ for Low Resistivity W", Journal of The Electrochemical Society, 2009, vol. 156, No. 9, pp. H685-H689.
- KR Office Action dated Aug. 1, 2022, in Application No. 10-2017-0172906 with English translation.
- KR Office Action dated Dec. 28, 2022 in Application No. KR10-2015-0137906 with English translation.
- KR Office Action dated Feb. 14, 2023 in Application No. KR10-2023-0013752 with English translation.
- KR Office Action dated Feb. 23, 2023 in Application No. KR10-2018-0044769 with English translation.
- KR Office Action dated Jan. 27, 2023 in Application No. KR10-2017-0102113 with English Translation.
- KR Office Action dated May 12, 2023 in Application No. KR10-2016-0060657 with English translation.
- KR Office Action dated Jun. 1, 2022, in Application No. 10-2017-0172906 with English translation.
- KR Office Action dated Sep. 27, 2022, in Application No. KR10-2017-0102113 with English translation.
- TW Office Action dated Jun. 28, 2023 in Application No. TW111120546 with English translation.
- U.S. Non-Final office Action dated Jan. 5, 2023 in U.S. Appl. No. 17/359,068.
- U.S. Restriction Requirement dated Sep. 15, 2022 in U.S. Appl. No. 17/359,068.
- U.S. Advisory Action dated Aug. 11, 2023, in U.S. Appl. No. 17/497,702.
- U.S. Final Office Action dated Jun. 28, 2023, in U.S. Appl. No. 17/359,068.
- U.S. Final Office Action dated May 16, 2023 in U.S. Appl. No. 17/497,702.
- U.S. Non-Final Office Action dated Feb. 2, 2023 in U.S. Appl. No. 17/497,702.
- U.S. Non-Final Office Action dated Jul. 7, 2023, in U.S. Appl. No. 17/809,480.
- U.S. Notice of Allowance dated Sep. 5, 2023, in U.S. Appl. No. 17/497,702.
- U.S. Appl. No. 17/907,377 inventors Khare et al., filed Sep. 26, 2022.
- U.S. Appl. No. 17/907,959, inventors Birru et al., filed Aug. 29, 2022.
- CN Office Action dated Jul. 26, 2024 in CN Application No. 202110914064.2, with English Translation.
- Coughlan J., et al., "Investigations of Stress Distributions in Tungsten-filled via Structures Using Finite Element Analysis," Microelectron. Reliab., 1997, vol. 37 (10/11), pp. 1549-1552.
- International Search Report and Written Opinion dated Apr. 20, 2023, in Application No. PCT/US2022/081047.
- International Search Report and Written Opinion dated Apr. 20, 2023 in PCT Application No. PCT/US2022/081398.
- Kang S., et al., "Application of Selective CVD Tungsten for Low Contact Resistance via Filling to Aluminum Multilayer Interconnection," Journal of Electronic Materials, 1988, vol. 17 (3), pp. 213-216.
- Kikuchi H., et al., "Tungsten Through-Silicon via Technology for Three-Dimensional LSIs," Japanese Journal of Applied Physics, 2008, vol. 47 (4), pp. 2801-2806.
- KR Office Action dated Jul. 19, 2024 in KR Application No. 10-2021-7021040 with English translation.
- KR Office Action dated Jun. 27, 2024 in KR Application No. 10-2023-0081299, with English Translation.
- U.S. Non-Final Office Action dated May 23, 2024 in U.S. Appl. No. 17/430,633.
- U.S. Appl. No. 18/715,602, inventor Tran S, filed May 31, 2024.
- U.S. Appl. No. 18/717,059, inventor Yan Z, filed Jun. 6, 2024.
- U.S. Restriction Requirement dated Jan. 29, 2024 in U.S. Appl. No. 17/430,633.
- CN Office Action dated Dec. 28, 2023 in CN Application No. 201980081000.X with English Translation.
- CN Office Action dated Oct. 11, 2023, in application No. CN202110914064.2 with English translation.
- CN Office Action dated Oct. 12, 2023, in application No. CN202110637340.5 with English translation.
- JP Office Action dated Nov. 28, 2023 in JP Application No. 2021-531900, with English Translation.
- KR Office Action dated Feb. 26, 2024 in KR Application No. 10-2015-0137906, with English Translation.
- KR Office Action dated Jul. 28, 2023, in Application No. KR10-2023-0081299 with English Translation.
- KR Office Action dated Oct. 26, 2023 in KR Application No. 10-2015-0137906, with English Translation.
- U.S. Corrected Notice of Allowance dated Jan. 10, 2024 in U.S. Appl. No. 17/497,702.
- U.S. Final Office Action dated Mar. 28, 2024 in U.S. Appl. No. 17/359,068.
- U.S. Non-Final Office Action dated Nov. 8, 2023 in U.S. Appl. No. 17/359,068.
- U.S. Notice of Allowance dated Apr. 3, 2024 in U.S. Appl. No. 17/299,753.
- U.S. Notice of Allowance dated Dec. 26, 2023 in U.S. Appl. No. 17/299,753.
- U.S. Notice of Allowance dated Feb. 14, 2024 in U.S. Appl. No. 17/299,753.
- U.S. Notice of Allowance dated Feb. 26, 2024 in U.S. Appl. No. 17/809,480.
- U.S. Notice of Allowance dated Jan. 8, 2024 in U.S. Appl. No. 17/299,753.
- U.S. Notice of Allowance dated Mar. 15, 2024 in U.S. Appl. No. 17/809,480.
- U.S. Appl. No. 18/616,741, inventors Chandrashekar A, et al., filed Mar. 26, 2024.
- U.S. Restriction Requirement dated Jun. 17, 2016 in U.S. Appl. No. 14/965,806.
- CN Office Action dated Dec. 17, 2024 in CN Application No. 202110914064.2, with English Translation.
- CN Office Action dated Feb. 27, 2025 in CN Application No. 202110914064.2, with English Translation.
- CN Office Action dated Mar. 25, 2025 in CN Application No. 201980081000.X, with English Translation.
- International Preliminary Report on Patentability and Written Opinion dated Nov. 30, 2023 in PCT Application No. PCT/US2022/030053.
- International Search Report and Written Opinion dated Sep. 8, 2022 in Application No. PCT/US2022/030053.
- JP Office Action dated Apr. 2, 2024 in JP Application No. 2021-547160, with English Translation.
- JP Office Action dated Apr. 15, 2025 in JP Application No. 20220552715, with English Translation.
- JP Office Action dated Jan. 7, 2025 in JP Application No. 2021-547160 with English translation.
- KR Notice of Allowance dated Feb. 6, 2025 in KR Application No. 10-2015-0137906, with English Translation.

(56)

References Cited

OTHER PUBLICATIONS

KR Office Action dated Apr. 23, 2025 in KR Application No. 10-2021-7029104, with English Translation.
 KR Office Action dated Jul. 27, 2024 in KR Application No. 10-2021-7029104, with English Translation.
 U.S. Corrected Notice of Allowance dated Feb. 20, 2025 in U.S. Appl. No. 17/430,633.
 U.S. Non-Final Office Action dated Dec. 19, 2024 in U.S. Appl. No. 17/809,480.
 U.S. Non-Final Office Action dated Feb. 11, 2025 in U.S. Appl. No. 18/616,741.
 U.S. Notice of Allowance dated Apr. 4, 2025 in U.S. Appl. No. 17/809,480.
 CN Office Action dated Aug. 21, 2024 in CN Application No. 202110637340.5 with English translation.
 CN Office Action dated Nov. 11, 2024 in CN Application No. 201980081000.X with English translation.
 JP Office Action dated Nov. 26, 2024 in JP Application No. 2021-531900, with English Translation.
 KR Notice of Allowance dated Nov. 25, 2024 in KR Application No. 10-2023-0081299, with English Translation.
 KR Office Action dated Sep. 30, 2024 in KR Application No. 10-2015-0137906 with English Translation.

TW Office Action dated Sep. 6, 2024 in TW Application No. 110107688 with English translation.
 U.S. Final Office Action dated Sep. 3, 2024 in U.S. Appl. No. 17/430,633.
 U.S. Final Office Action dated Sep. 30, 2024 in U.S. Appl. No. 17/809,480.
 U.S. Notice of Allowance dated Nov. 14, 2024 in U.S. Appl. No. 17/430,633.
 U.S. Appl. No. 18/943,613, inventors Chandrashekar A, et al., filed Nov. 11, 2024.
 U.S. Supplemental Notice of Allowance dated Nov. 20, 2024 in U.S. Appl. No. 17/430,633.
 International Preliminary Report on Patentability and Written Opinion dated Jun. 20, 2024 in PCT Application No. PCT/US2022/081047.
 International Preliminary Report on Patentability and Written Opinion dated Jun. 27, 2024 in PCT Application No. PCT/US2022/081398.
 JP Office Action dated May 28, 2024 in JP Application No. 2021-531900, with English Translation.
 U.S. Non-Final Office Action dated Jun. 20, 2024 in U.S. Appl. No. 17/809,480.

* cited by examiner

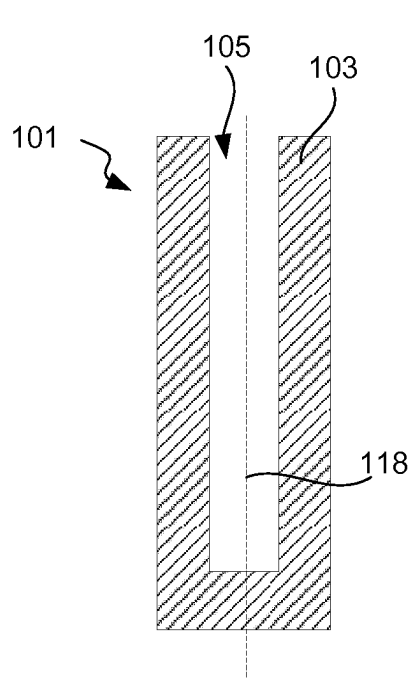


FIG. 1A

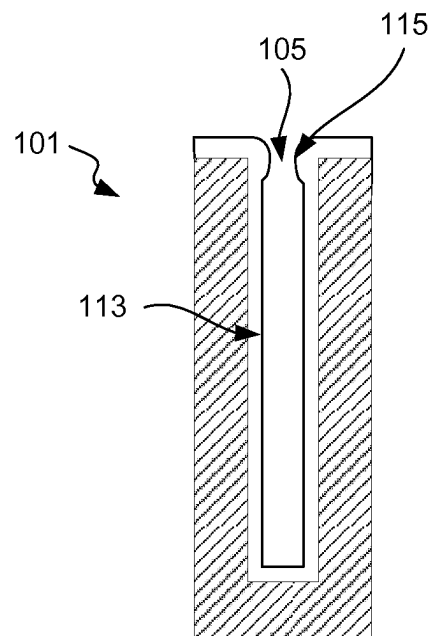


FIG. 1B

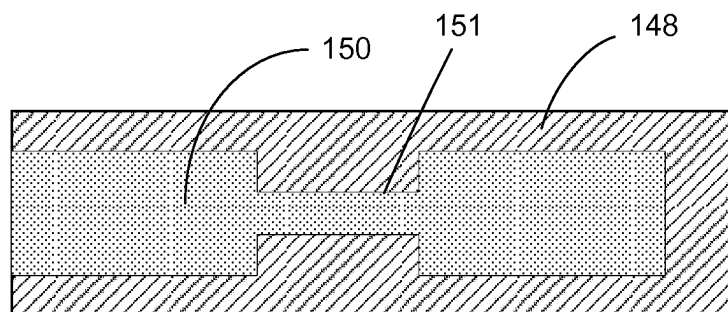


FIG. 1D

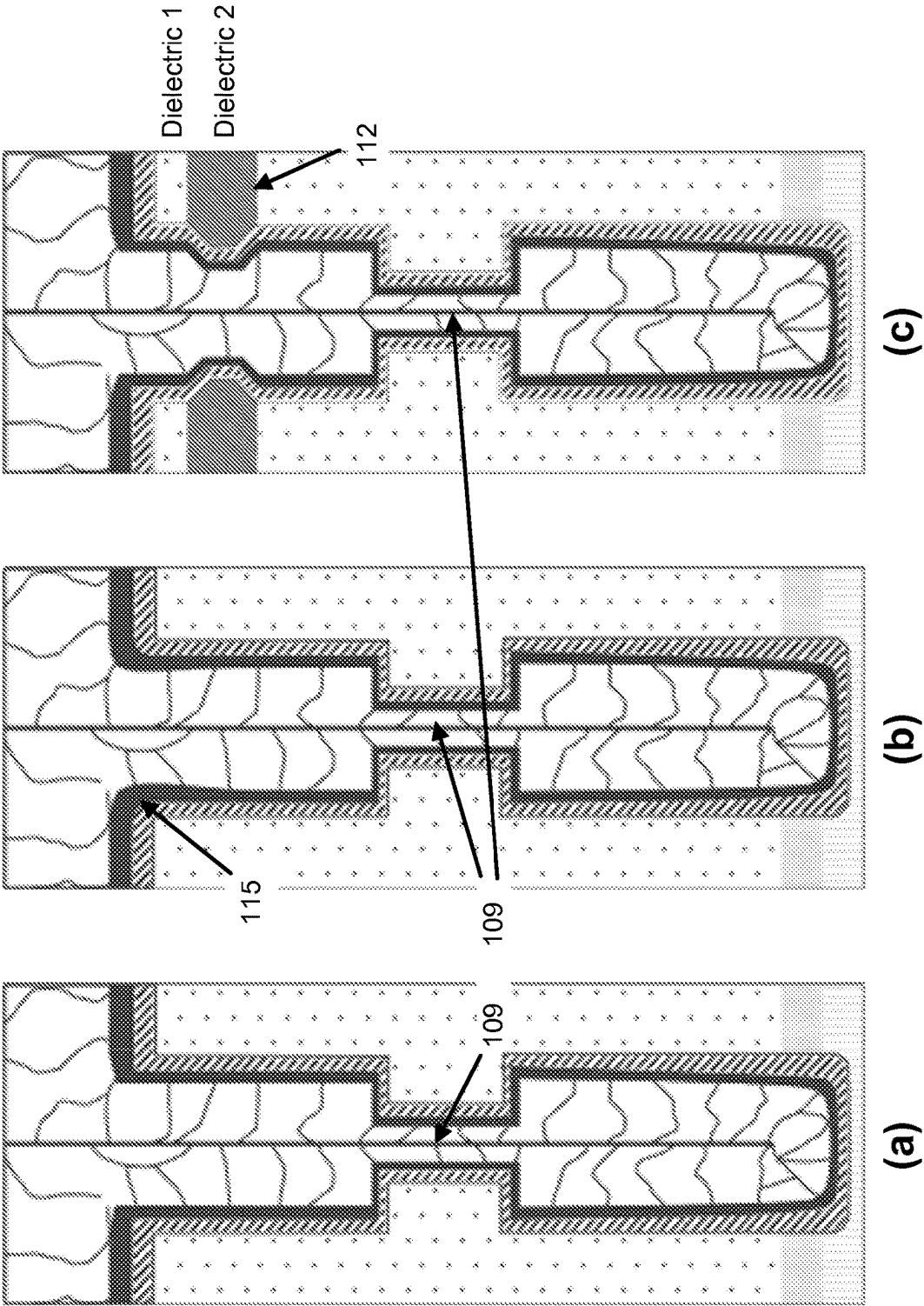


FIGURE 1C

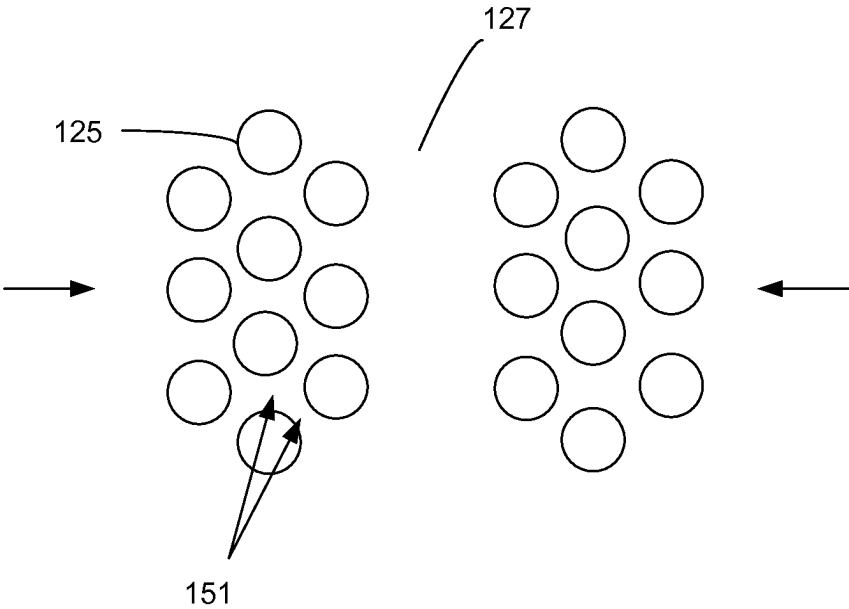


FIG. 1E

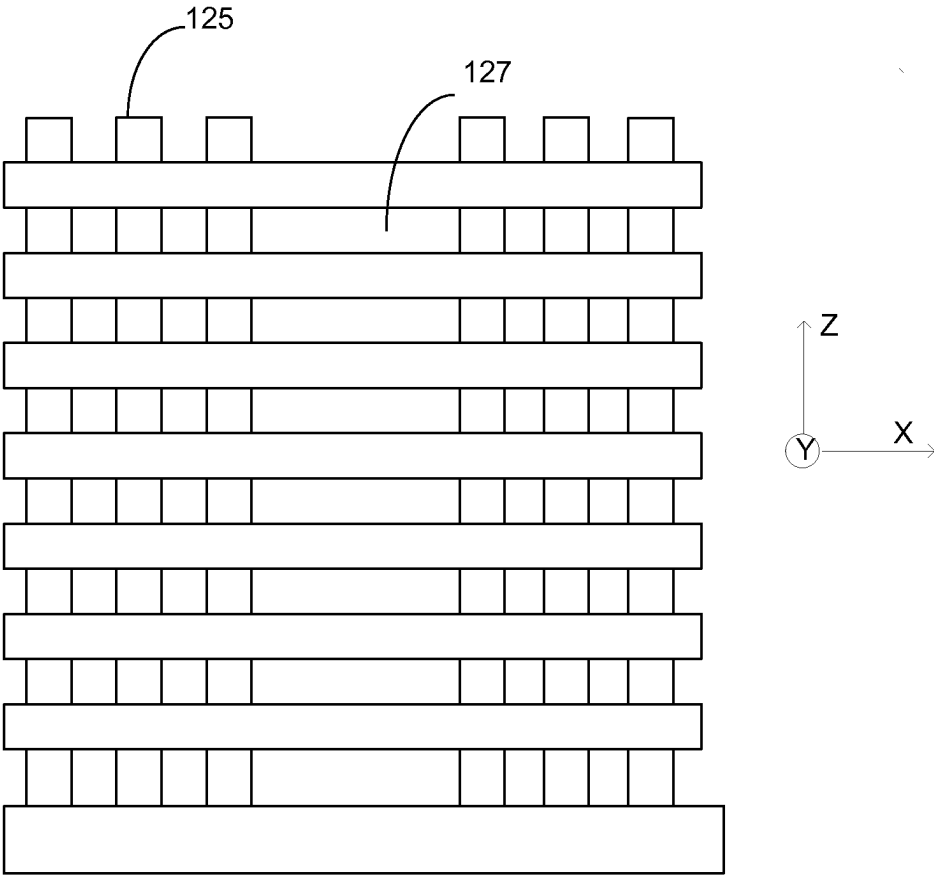


FIG. 1F

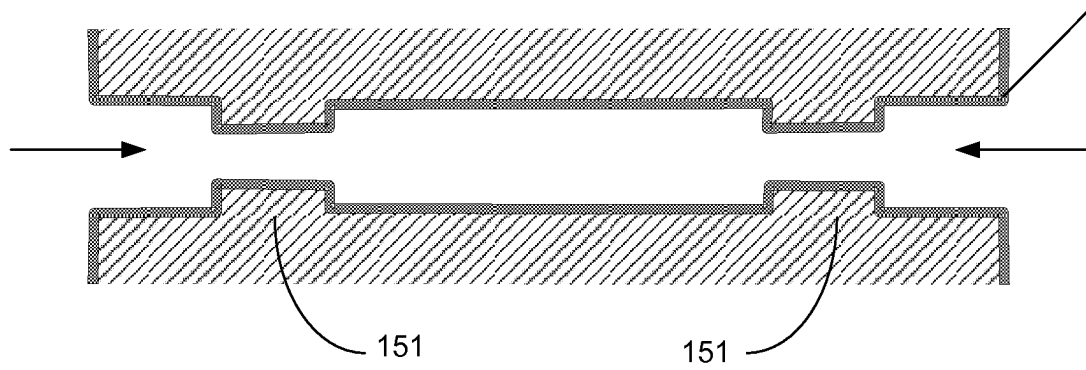


FIG. 1G

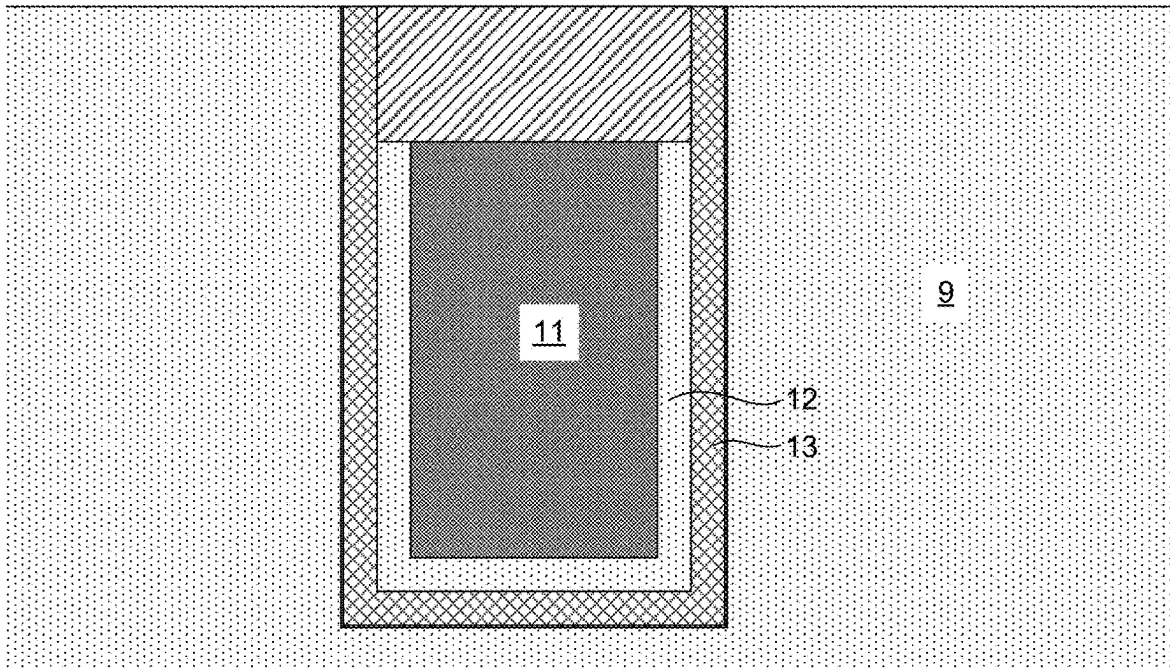
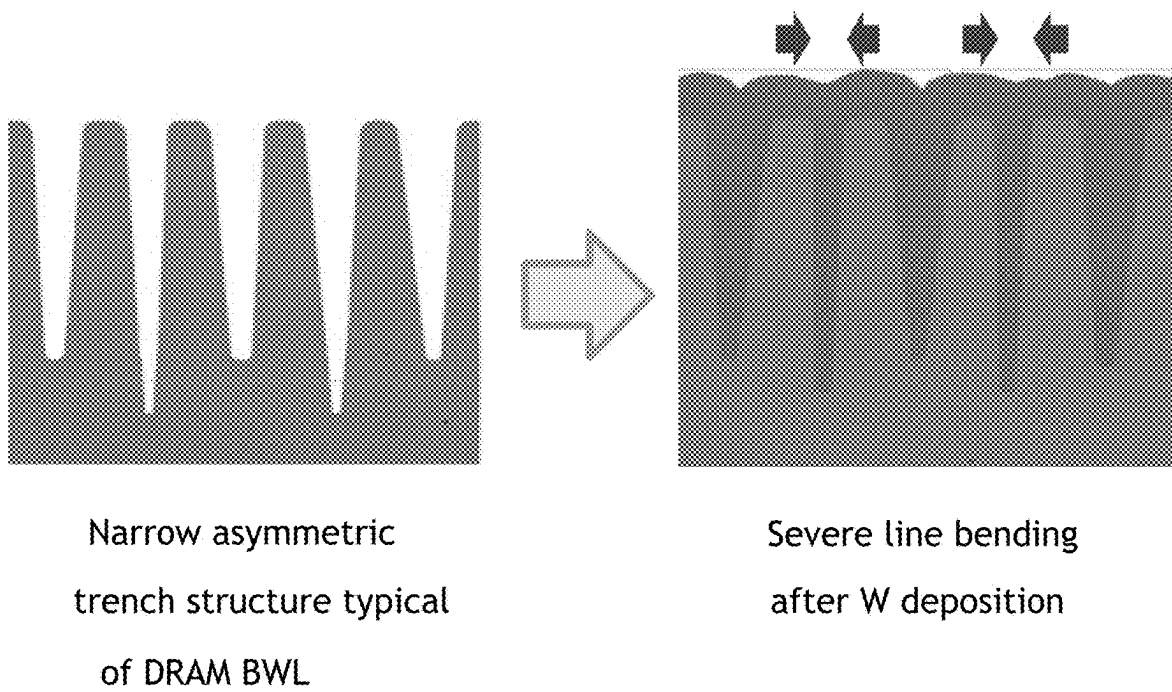


FIG. 1H

**FIG. 1I**

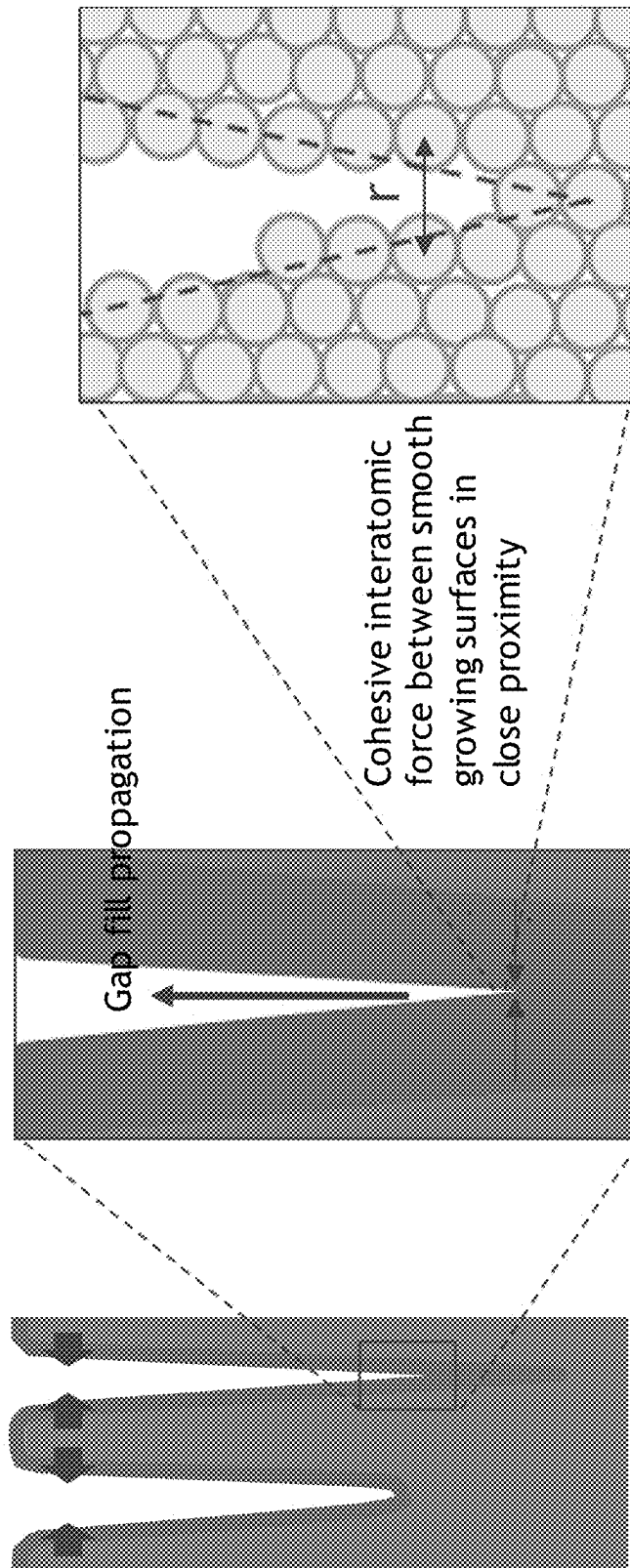
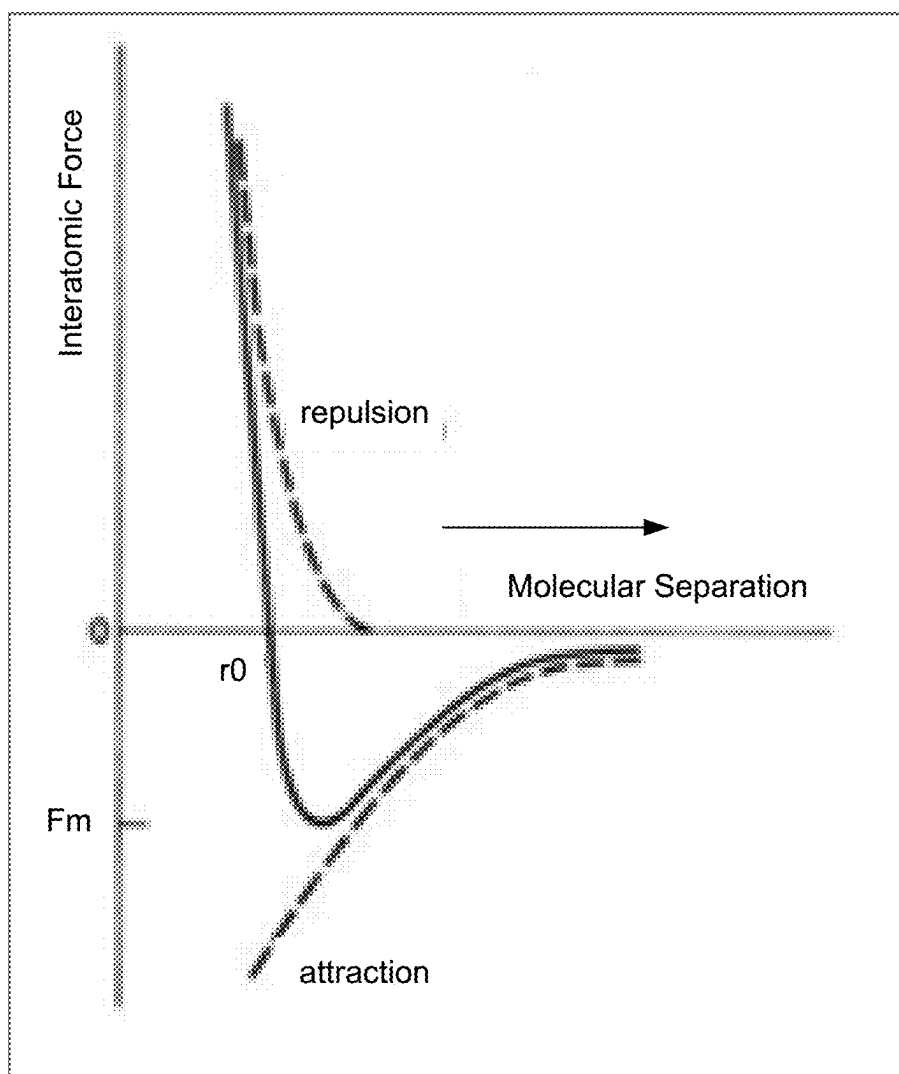
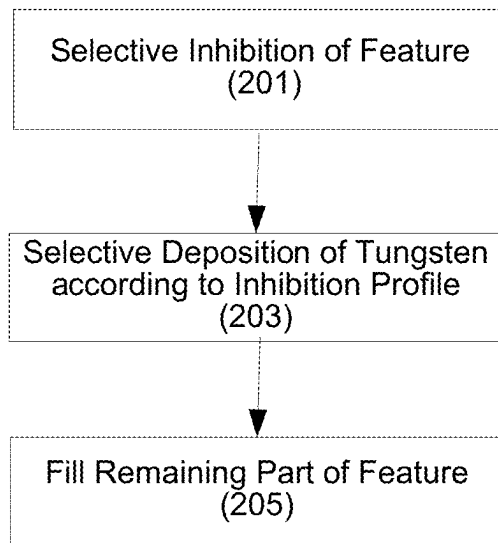
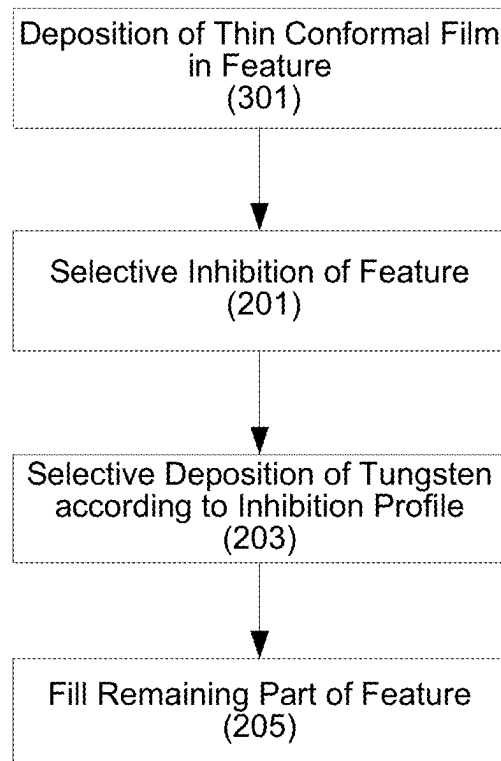
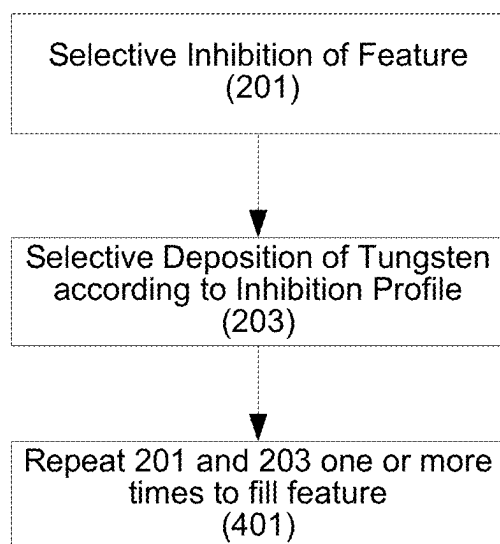


FIG. 1J

**FIG. 1K**

**FIG. 2**

**FIG. 3**

**FIG. 4**

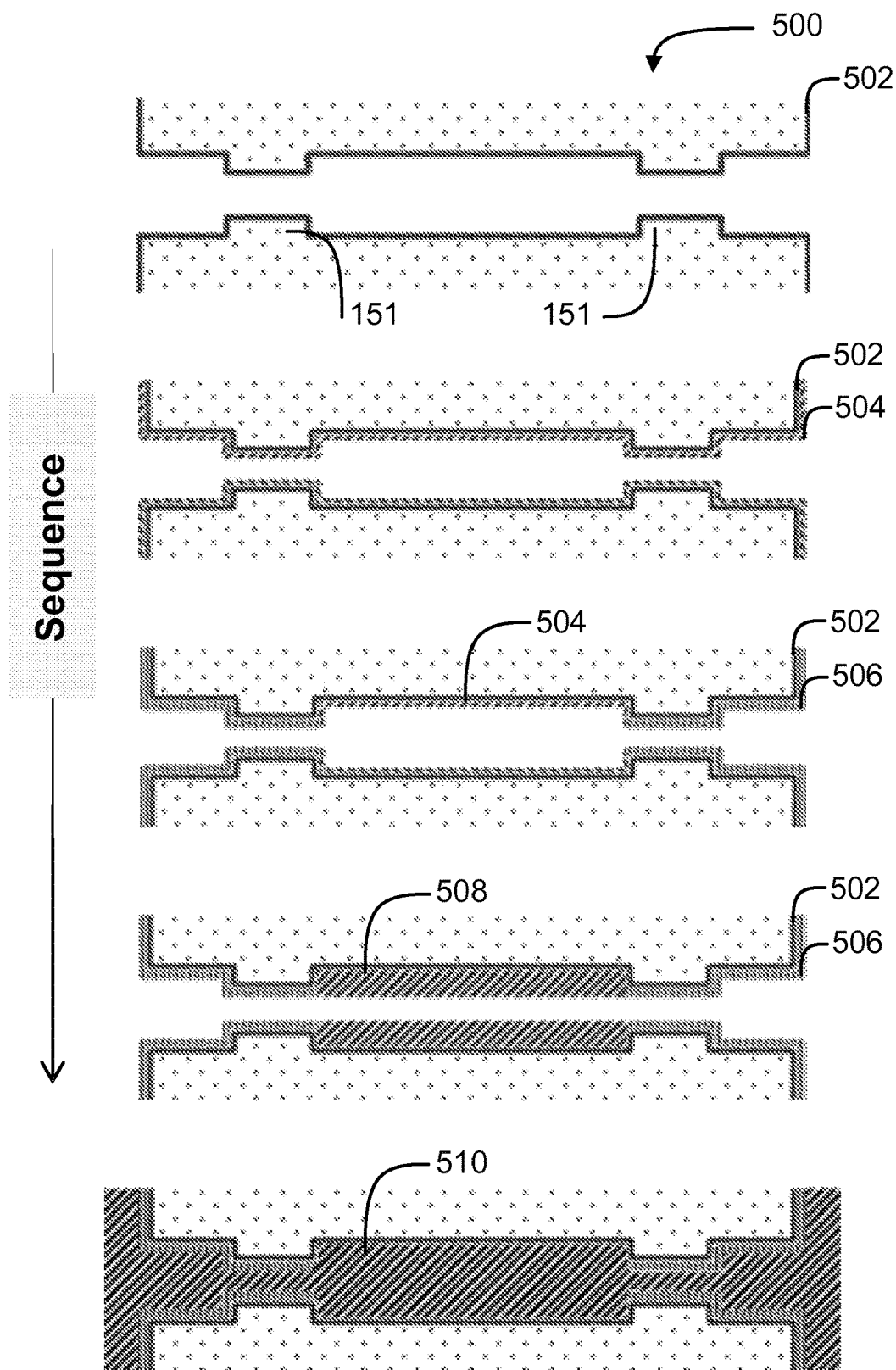


FIGURE 5

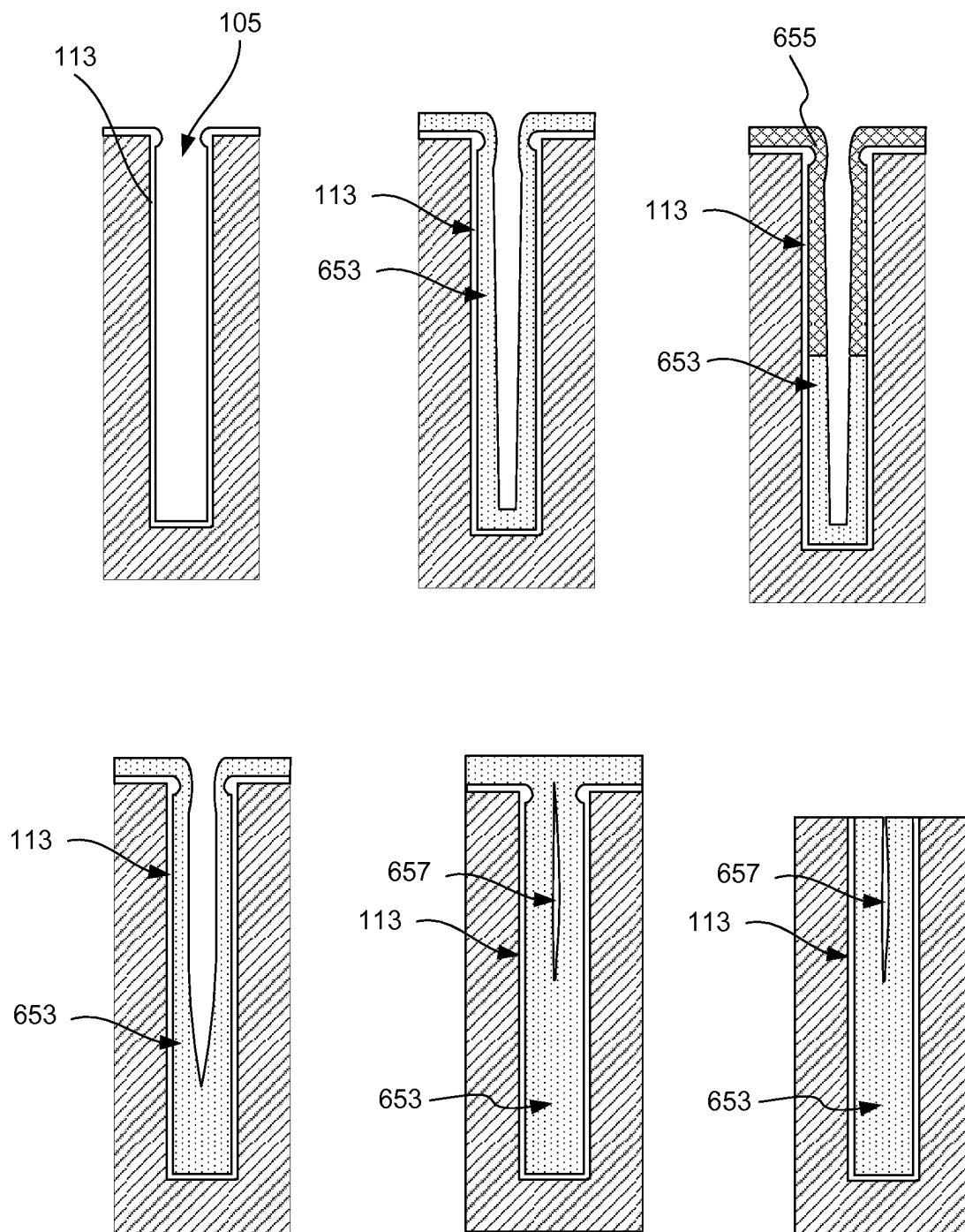


FIG. 6

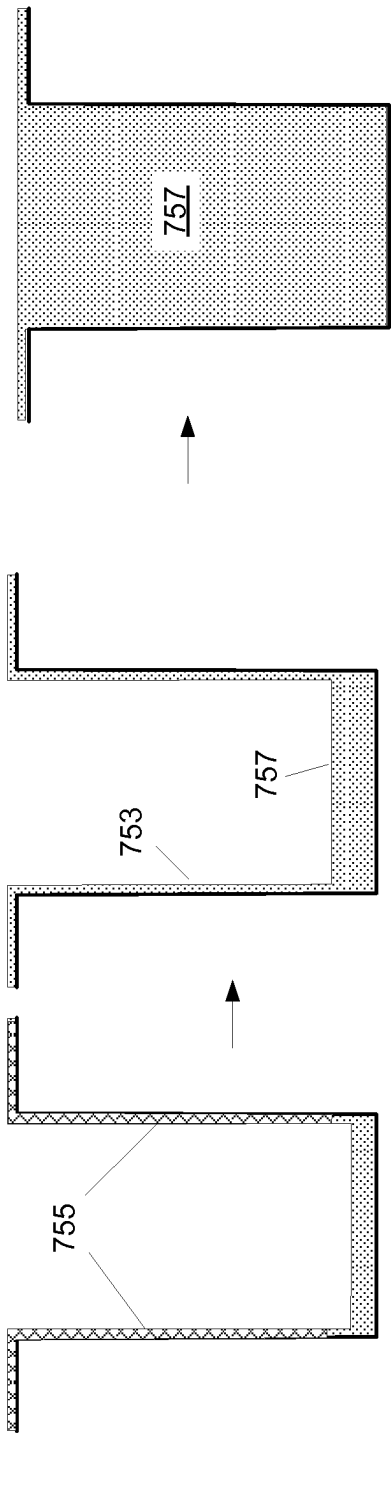
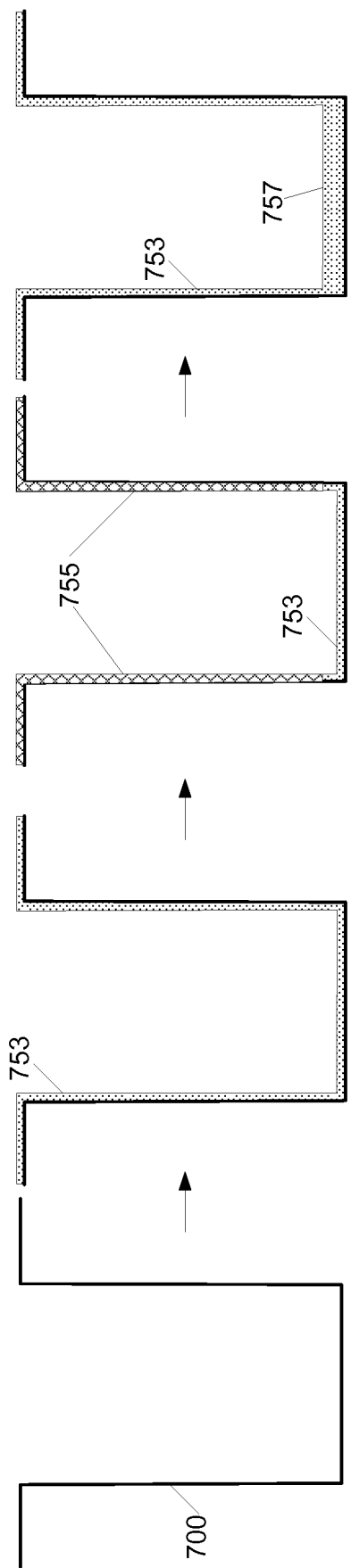


FIG. 7

repeat

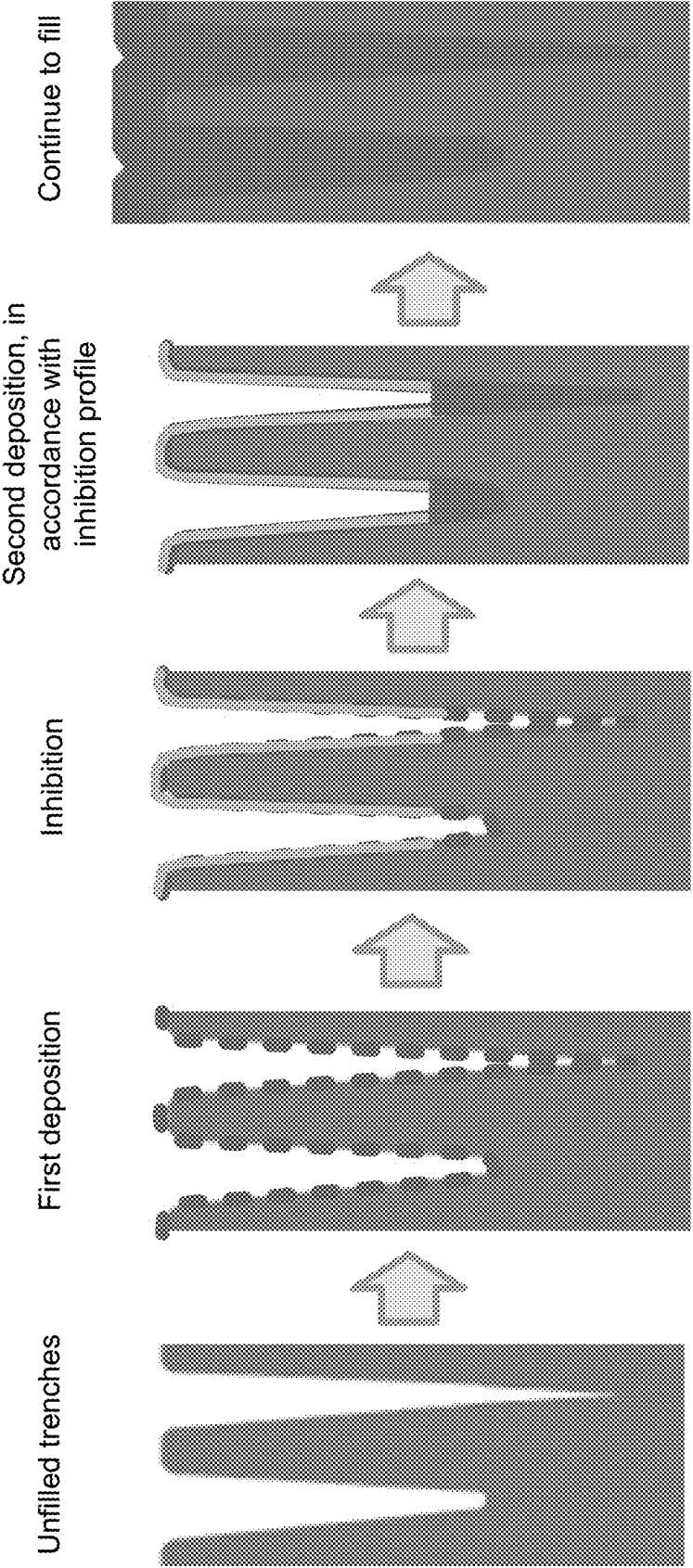
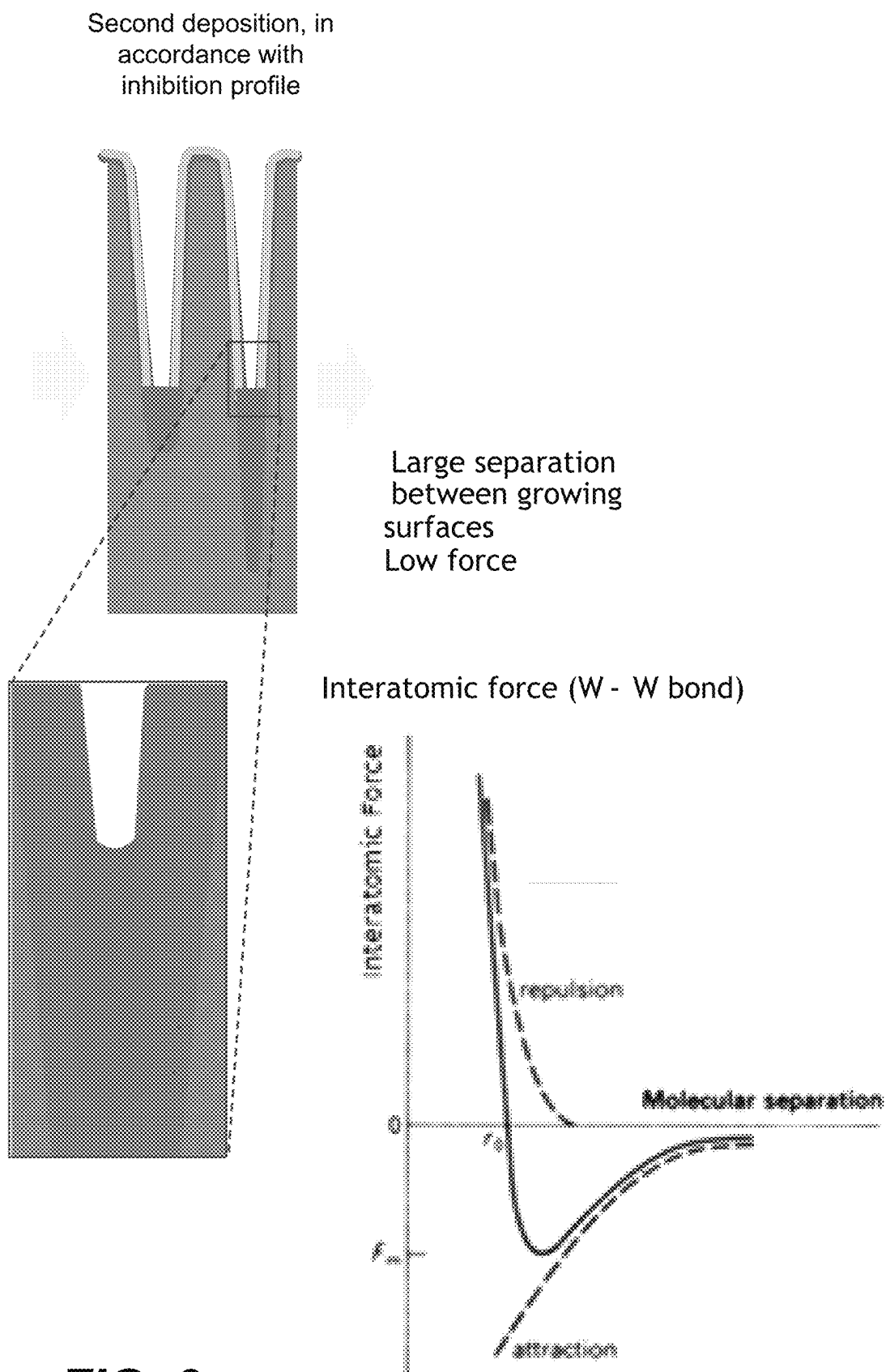


FIG. 8

**FIG. 9**

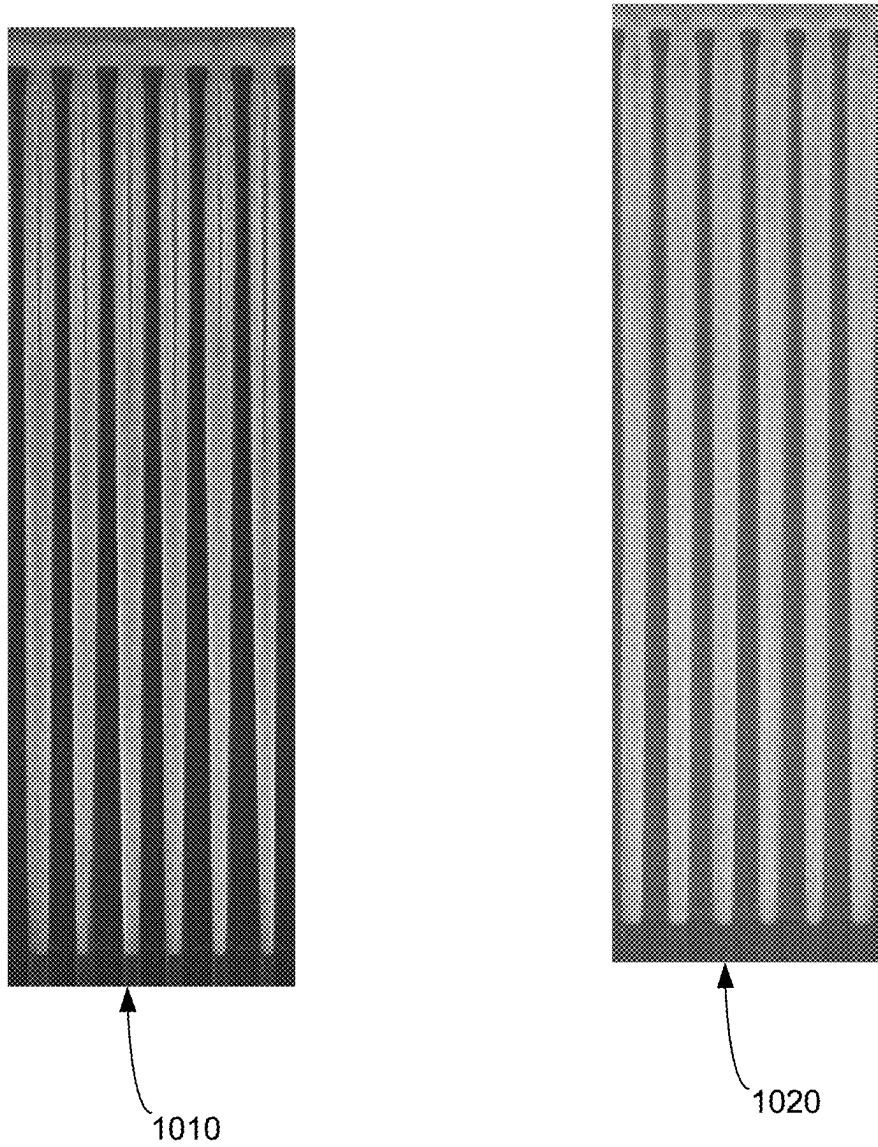


FIG. 10A

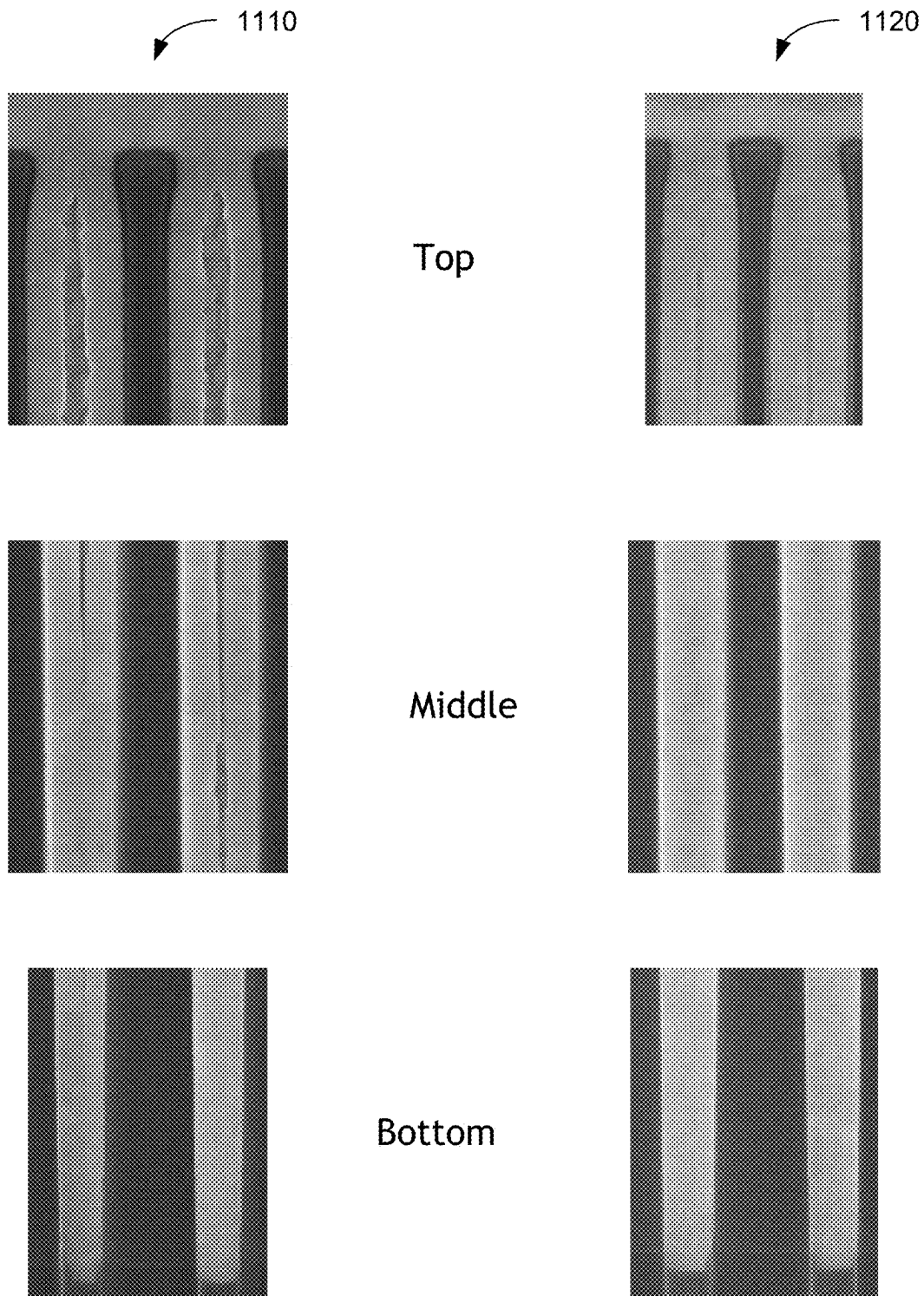
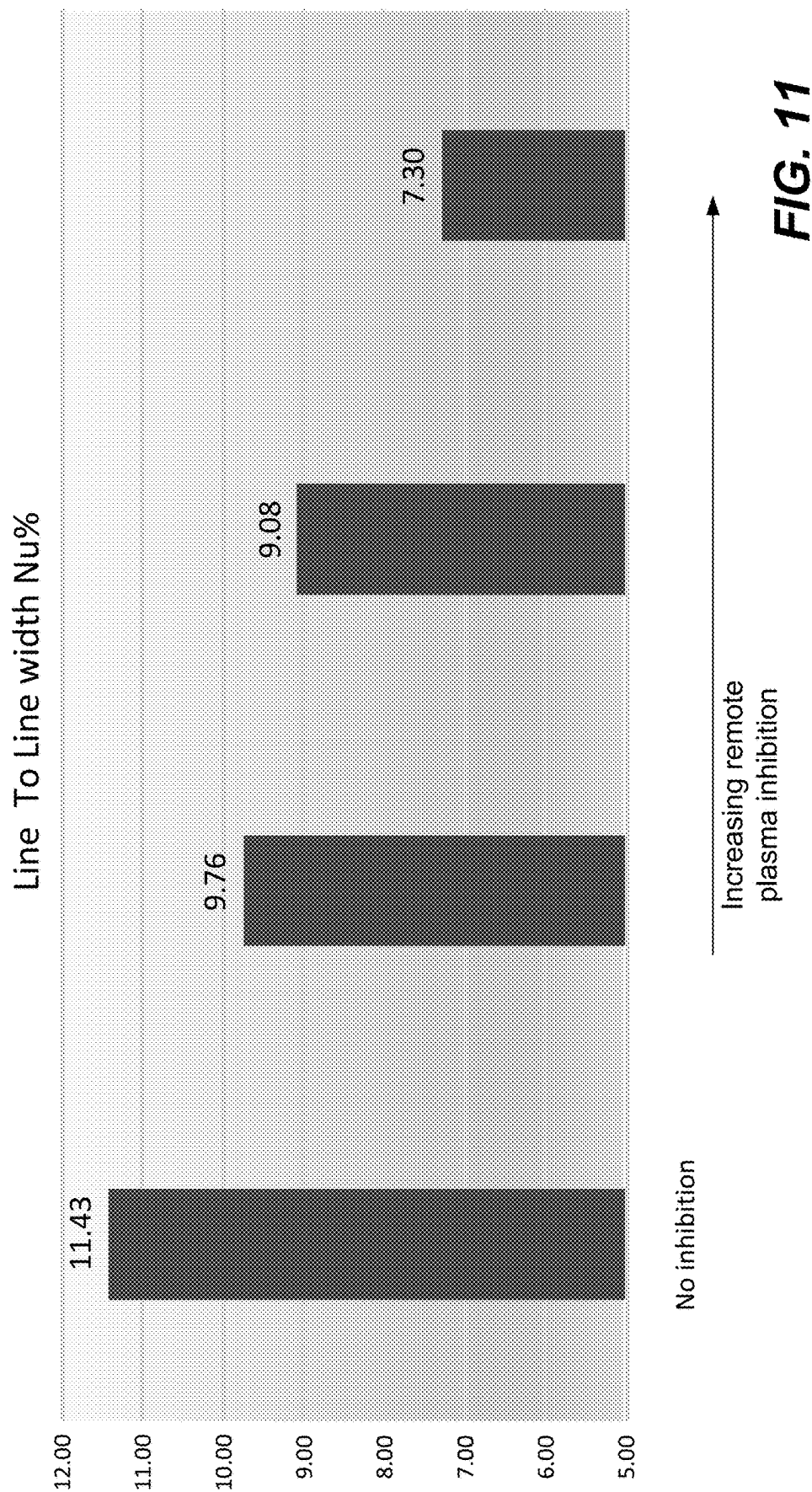
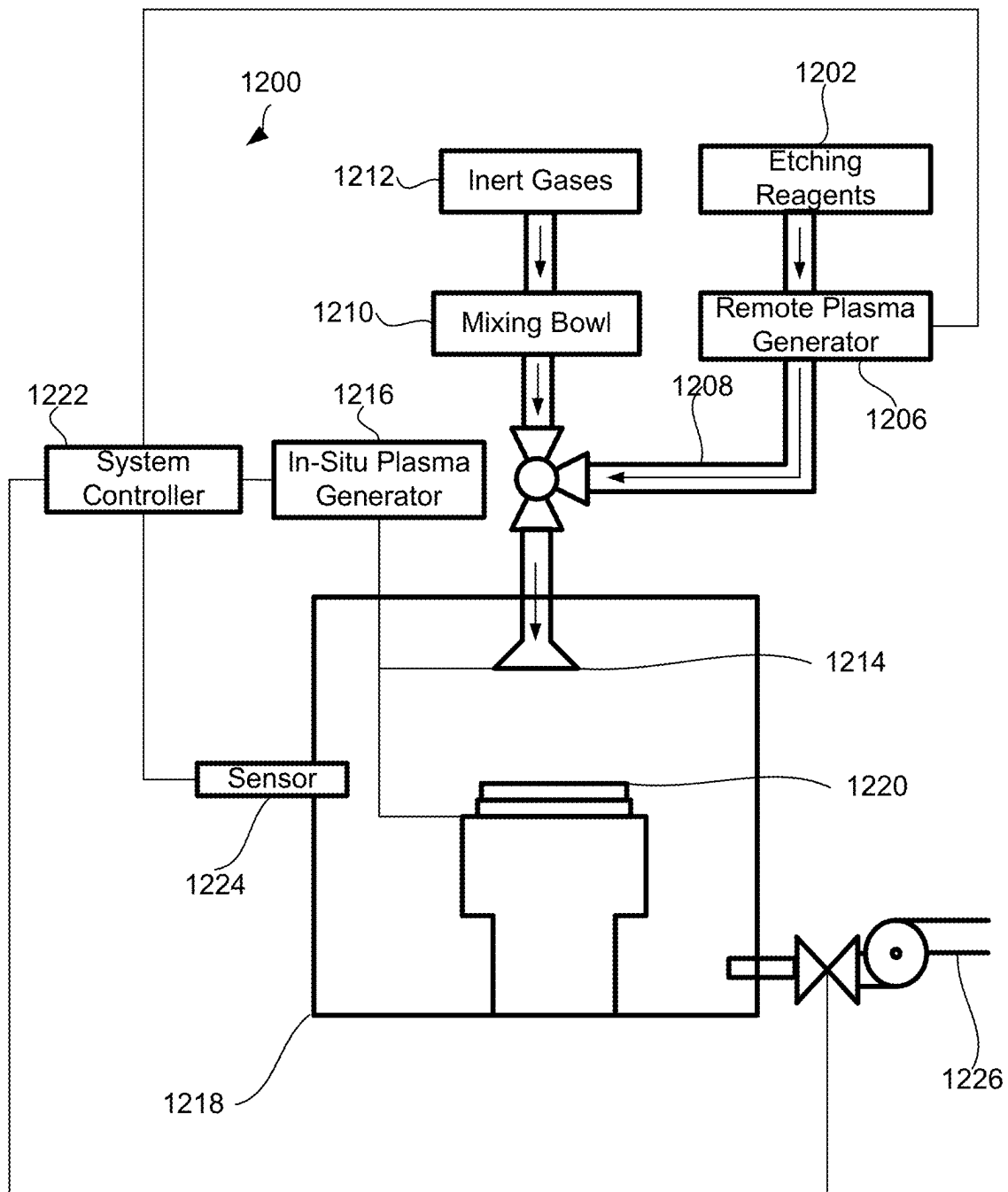


FIG. 10B



**FIG. 12**

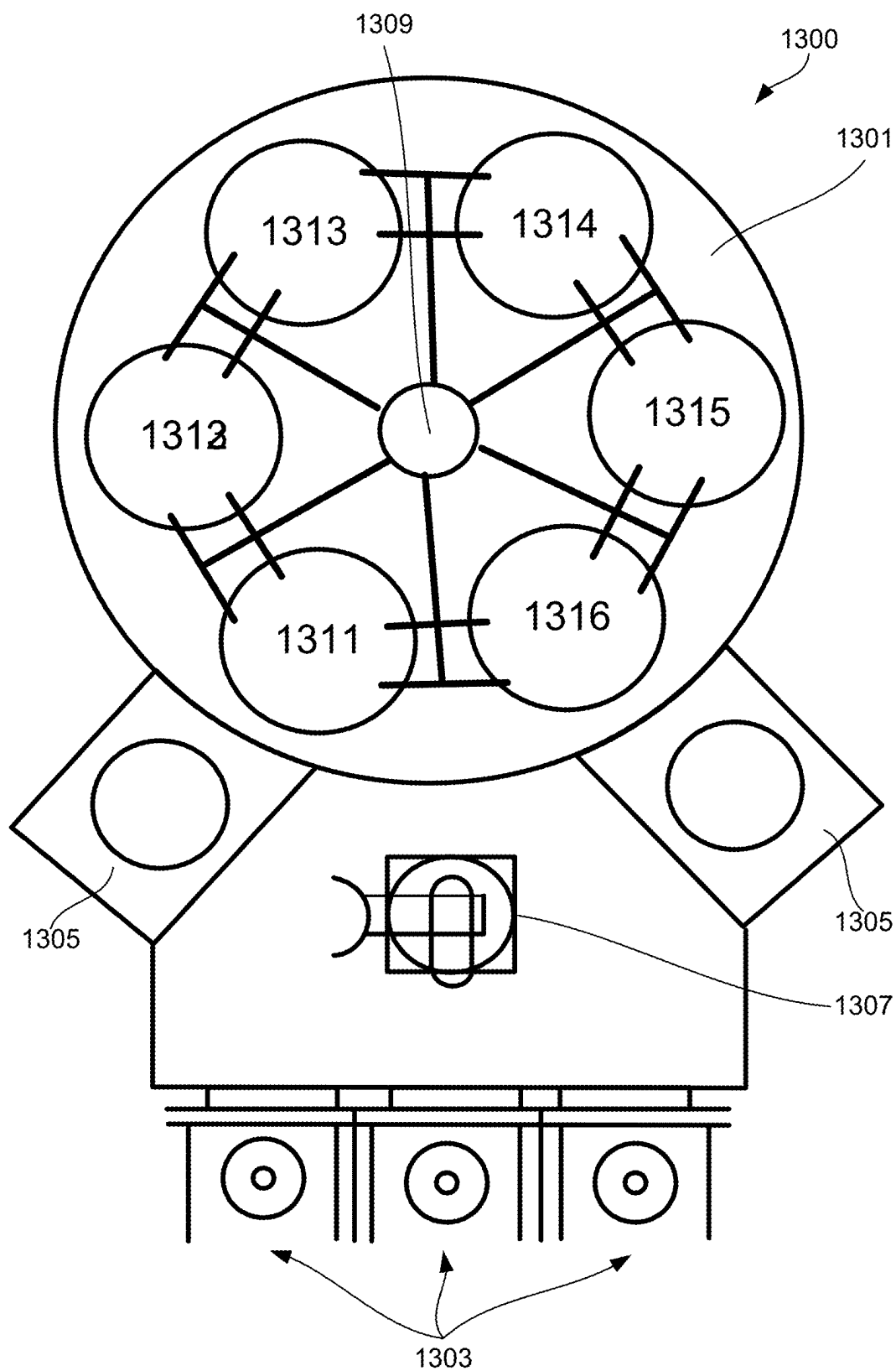


FIG. 13A

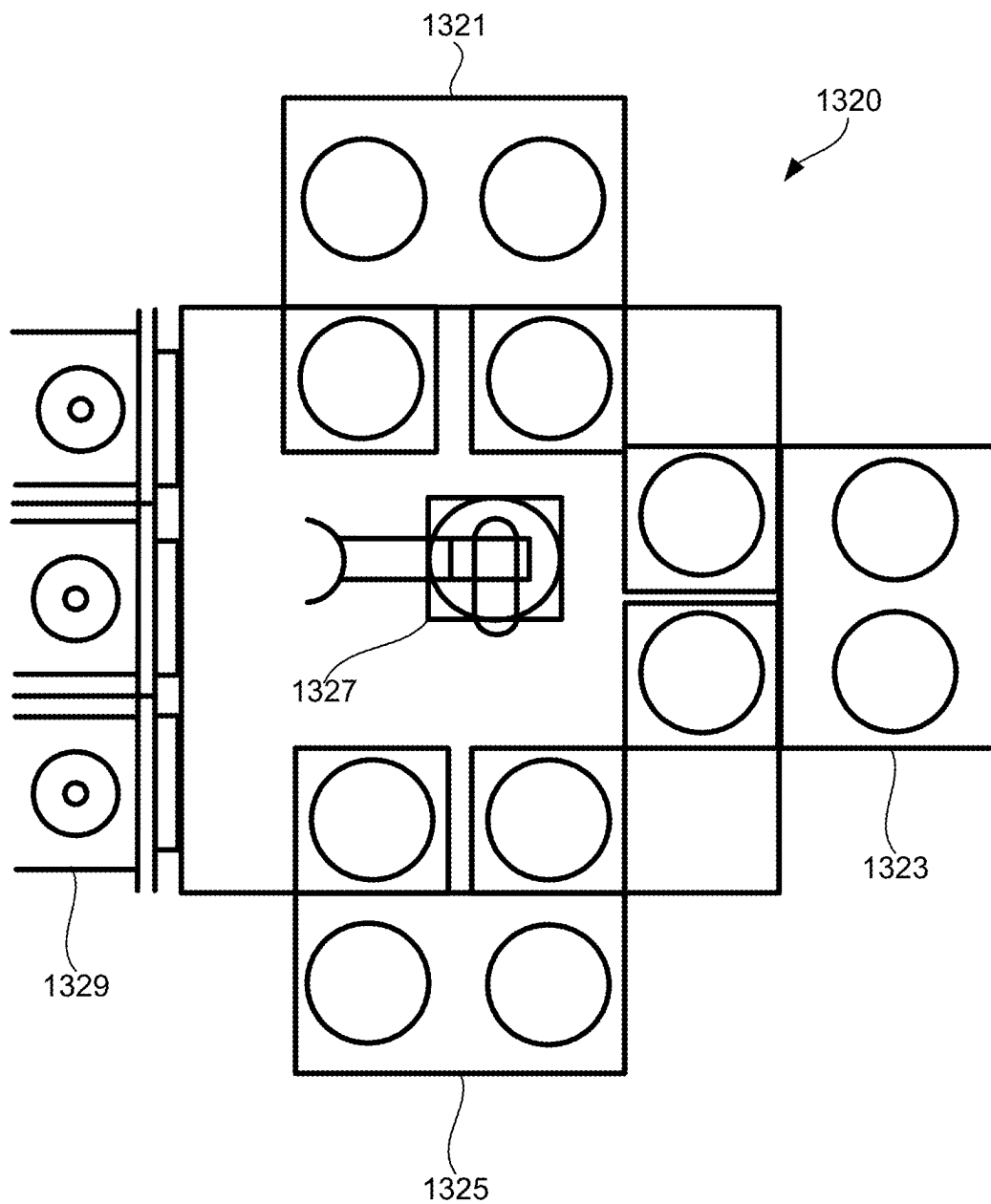


FIG. 13B

1

TUNGSTEN FEATURE FILL WITH NUCLEATION INHIBITION

INCORPORATION BY REFERENCE

An Application Data Sheet is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed Application Data Sheet is incorporated by reference herein in its entirety and for all purposes.

BACKGROUND

Deposition of conductive materials using chemical vapor deposition (CVD) techniques is an integral part of many semiconductor fabrication processes. These materials may be used for horizontal interconnects, vias between adjacent metal layers, contacts between first metal layers and devices on the silicon substrate, and high aspect ratio features. In a conventional tungsten deposition process, a substrate is heated to a predetermined process temperature in a deposition chamber, and a thin layer of tungsten-containing materials that serves as a seed or nucleation layer is deposited. Thereafter, the remainder of the tungsten-containing material (the bulk layer) is deposited on the nucleation layer. Conventionally, the tungsten-containing materials are formed by the reduction of tungsten hexafluoride (WF_6) with hydrogen (H_2). Tungsten-containing materials are deposited over an entire exposed surface area of the substrate including features and a field region.

Depositing tungsten-containing materials into small and, especially, high aspect ratio features may cause formation of seams and voids inside the filled features. Large seams may lead to high resistance, contamination, loss of filled materials, and otherwise degrade performance of integrated circuits. For example, a seam may extend close to the field region after filling process and then open during chemical-mechanical planarization.

SUMMARY

One aspect of the disclosure relates to a method including providing a substrate including a feature having one or more feature openings and a feature interior; selectively inhibiting tungsten nucleation in the feature such that there is a differential inhibition profile along a feature axis by exposing the feature to ammonia vapor in a non-plasma process; and selectively depositing tungsten in the feature in accordance with the differential inhibition profile. In some embodiments, selectively inhibiting tungsten nucleation in the feature further involves exposing the feature to a reducing agent and a tungsten-containing precursor. In some embodiments, the method involves depositing a tungsten layer in the feature prior to selective inhibition. The tungsten layer may be deposited by a pulsed nucleation layer (PNL) process. In some embodiments, the tungsten layer is conformally deposited in the feature. In some embodiments, selectively depositing tungsten involves chemical vapor deposition (CVD) process. In some embodiments, the method involves, after selectively depositing tungsten in the feature, depositing tungsten in the feature to complete feature fill. In some embodiments, the method involves, after selectively depositing tungsten in the feature, non-selectively depositing tungsten in the feature. In some embodiments, selectively inhibiting tungsten nucleation involves treating a tungsten surface of the feature. In some embodi-

2

ments, selectively inhibiting tungsten nucleation involves forming a tungsten nitride surface in the feature.

Another aspect of the disclosure relates to a method of filling multiple adjacent trenches with a metal. The method involves providing a substrate including multiple adjacent trenches, depositing a conformal layer of metal in the multiple adjacent trenches, selectively inhibiting nucleation on the conformal layer of metal at the top of the multiple adjacent trenches with respect to the bottom of the multiple adjacent trenches, and depositing metal at the bottom of the multiple adjacent trenches while preventing metal from growing from the sidewalls of each of the multiple adjacent trenches to thereby reduce line-to-line non-uniformity.

In some embodiments, the metal is tungsten. In some embodiments, selectively inhibiting nucleation on a conformal layer of tungsten is a remote plasma process. In some embodiments, the remote plasma process involves exposing the feature to nitrogen radicals.

In some embodiments, inhibiting tungsten nucleation on the conformal layer of tungsten is a thermal process. In some such embodiments, selectively inhibiting tungsten nucleation on the conformal layer of tungsten involves exposing the layer to ammonia vapor.

These and other aspects are described further below with reference to the Figures.

BRIEF DESCRIPTION OF DRAWINGS

FIGS. 1A-1G show examples of various structures that can be filled according to the processes described herein.

FIG. 1H depicts a schematic example of a DRAM architecture including a buried wordline (bWL) in a silicon substrate.

FIG. 1I shows an unfilled and filled narrow asymmetric trench structure typical of DRAM bWL.

FIG. 1J illustrates the phenomena of line bending during gap fill.

FIG. 1K is a graph illustrating interatomic force as a function of tungsten-tungsten bond radius, r .

FIGS. 2-4 are process flow diagrams illustrating certain operations in methods of filling features with tungsten.

FIGS. 5-7 are schematic diagrams showing features at various stages of feature fill.

FIG. 8 shows an example of trenches filled according to a method as shown in FIG. 3.

FIG. 9 shows an example of a deposition step in which tungsten is deposited in accordance with an inhibition profile.

FIG. 10A shows images of fill improvement from a thermal (no plasma) inhibition process.

FIG. 10B shows side-by-side enlarged comparisons of feature fill with no inhibition and thermal inhibition at the top, middle, and bottom of two features from the images in FIG. 10A.

FIG. 11 is a chart showing the effect of remote plasma inhibition on line-to-line non-uniformity.

FIGS. 12, 13A, and 13B are schematic diagrams showing examples of apparatus suitable for practicing the methods described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. The present invention may be practiced without some or all of these specific details. In other

3

instances, well known process operations have not been described in detail to not unnecessarily obscure the present invention. While the invention will be described in conjunction with the specific embodiments, it will be understood that it is not intended to limit the invention to the embodiments.

Described herein are methods of filling features with tungsten and related systems and apparatus. Examples of application include logic and memory contact fill, DRAM buried wordline fill, vertically integrated memory gate/wordline fill, and 3-D integration with through-silicon vias (TSVs). The methods described herein can be used to fill vertical features, such as in tungsten vias, and horizontal features, such as vertical NAND (VNAND) wordlines. The methods may be used for conformal and bottom-up or inside-out fill.

According to various embodiments, the features can be characterized by one or more of narrow and/or re-entrant openings, constrictions within the feature, and high aspect ratios. Examples of features that can be filled are depicted in FIGS. 1A-1C. FIG. 1A shows an example of a cross-sectional depiction of a vertical feature **101** to be filled with tungsten. The feature can include a feature hole **105** in a substrate **103**. The substrate may be a silicon wafer, e.g., 200-mm wafer, 300-mm wafer, 450-mm wafer, including wafers having one or more layers of material such as dielectric, conducting, or semi-conducting material deposited thereon. In some embodiments, the feature hole **105** may have an aspect ratio of at least about 2:1, at least about 4:1, at least about 6:1 or higher. The feature hole **105** may also have a dimension near the opening, e.g., an opening diameter or line width, of between about 10 nm to 500 nm, for example between about 25 nm to 300 nm. The feature hole **105** can be referred to as an unfilled feature or simply a feature. The feature, and any feature, may be characterized in part by an axis **118** that extends through the length of the feature, with vertically-oriented features having vertical axes and horizontally-oriented features having horizontal axes.

FIG. 1B shows an example of a feature **101** that has a re-entrant profile. A re-entrant profile is a profile that narrows from a bottom, closed end, or interior of the feature to the feature opening. According to various embodiments, the profile may narrow gradually and/or include an overhang at the feature opening. FIG. 1B shows an example of the latter, with an under-layer **113** lining the sidewall or interior surfaces of the feature hole **105**. The under-layer **113** can be for example, a diffusion barrier layer, an adhesion layer, a nucleation layer, a combination of thereof, or any other applicable material. The under-layer **113** forms an overhang **115** such that the under-layer **113** is thicker near the opening of the feature **101** than inside the feature **101**.

In some embodiments, features having one or more constrictions within the feature may be filled. FIG. 1C shows examples of views of various filled features having constrictions. Each of the examples (a), (b) and (c) in FIG. 1C includes a constriction **109** at a midpoint within the feature. The constriction **109** can be, for example, between about 15 nm-20 nm wide. Constrictions can cause pinch off during deposition of tungsten in the feature using conventional techniques, with deposited tungsten blocking further deposition past the constriction before that portion of the feature is filled, resulting in voids in the feature. Example (b) further includes a liner/barrier overhang **115** at the feature opening. Such an overhang could also be a potential pinch-off point. Example (c) includes a constriction **112** further away from the field region than the overhang **115** in example (b). As

4

described further below, methods described herein allow void-free fill as depicted in FIG. 1C.

Horizontal features, such as in 3-D memory structures, can also be filled. FIG. 1D shows an example of a word line **150** in a VNAND structure **148** that includes a constriction **151**. In some embodiments, the constrictions can be due to the presence of pillars in a VNAND or other structure. FIG. 1E, for example, shows a plan view of pillars **125** in a VNAND structure, with FIG. 1F showing a simplified schematic of a cross-sectional depiction of the pillars **125**. Arrows in FIG. 1E represent deposition material; as pillars **125** are disposed between an area **127** and a gas inlet or other deposition source, adjacent pillars can result in constrictions that present challenges in void free fill of an area **127**.

FIG. 1G provides another example of a view horizontal feature, for example, of a VNAND or other structure including pillar constrictions **151**. The example in FIG. 1G is open-ended, with material to be deposited able to enter laterally from two sides as indicated by the arrows. (It should be noted that example in FIG. 1G can be seen as a 2-D rendering of 3-D features of the structure, with the FIG. 1G being a cross-sectional depiction of an area to be filled and pillar constrictions shown in the figure representing constrictions that would be seen in a plan rather than cross-sectional view.) In some embodiments, 3-D structures can be characterized with the area to be filled extending along three dimensions (e.g., in the X, Y and Z-directions in the example of FIG. 1F), and can present more challenges for fill than filling holes or trenches that extend along one or two dimensions. For example, controlling fill of a 3-D structure can be challenging as deposition gasses may enter a feature from multiple dimensions.

Filling features with tungsten-containing materials may cause formation of voids and seams inside the filled features. A void is region in the feature that is left unfilled. A void can form, for example, when the deposited material forms a pinch point within the feature, sealing off an unfilled space within the feature preventing reactant entry and deposition.

There are multiple potential causes for void and seam formation. One is an overhang formed near the feature opening during deposition of tungsten-containing materials or, more typically, other materials, such as a diffusion barrier layer or a nucleation layer. An example is shown in FIG. 1B.

Another cause of void or seam formation that is not illustrated in FIG. 1B but that nevertheless may lead to seam formation or enlarging seams is curved (or bowed) side walls of feature holes, which are also referred to as bowed features. In a bowed feature the cross-sectional dimension of the cavity near the opening is smaller than that inside the feature. Effects of these narrower openings in the bowed features are somewhat similar to the overhang problem described above. Constrictions within a feature such as shown in FIGS. 1C, 1D and 1G also present challenges for tungsten fill without few or no voids and seams.

Even if void free fill is achieved, tungsten in the feature may contain a seam running through the axis or middle of the via, trench, line or other feature. This is because tungsten growth can begin at the sidewall and continues until the grains meet with tungsten growing from the opposite sidewall. This seam can allow for trapping of impurities including fluorine-containing compounds such as hydrofluoric acid (HF). During chemical mechanical planarization (CMP), coring can also propagate from the seam. According to various embodiments, the methods described herein can reduce or eliminate void and seam formation. The methods described herein may also address one or more of the following:

- 1) Very challenging profiles: Void free fill can be achieved in most re-entrant features using dep-etch-dep cycles as described in U.S. patent application Ser. No. 13/351,970, incorporated by reference herein. However, depending on the dimensions and geometry, multiple dep-etch cycles may be needed to achieve void-free fill. This can affect process stability and throughput. Embodiments described herein can provide feature fill with fewer or no dep-etch-dep cycles.
- 2) Small features and liner/barrier impact: In cases where the feature sizes are extremely small, tuning the etch process without impacting the integrity of the under-layer liner/barrier can be very difficult. In some cases intermittent Ti attack—possibly due to formation of a passivating TiFx layer during the etch—can occur during a W-selective etch.
- 3) Scattering at W grain boundaries: Presence of multiple W grains inside the feature can result in electron loss due to grain boundary scattering. As a result, actual device performance will be degraded compared to theoretical predictions and blanket wafer results.
- 4) Reduced via volume for W fill: Especially in smaller and newer features, a significant part of the metal contact is used up by the W barrier (TiN, WN etc.). These films are typically higher resistivity than W and negatively impact electrical characteristics like contact resistance etc.

Particular embodiments relate to methods and related apparatus for formation of tungsten wordlines in memory devices. FIG. 1H depicts a schematic example of a DRAM architecture including a buried wordline (bWL) **11** in a silicon substrate **9**. The bWL is formed in a trench etched in the silicon substrate **9**. Lining the trench is a conformal barrier layer **12** and an insulating layer **13** that is disposed between the conformal barrier layer **12** and the silicon substrate **9**. In the example of FIG. 1H, the insulating layer **13** may be a gate oxide layer, formed from a high-k dielectric material such as a silicon oxide or silicon nitride material. Examples of conformal barrier layers include tungsten-containing layers and titanium nitride (TiN). Tungsten-containing conformal barrier layers are described in U.S. patent application Ser. No. 15/051,561, which is incorporated by reference herein.

Conventional deposition processes for DRAM bWL trenches tend to distort the trenches such that the final trench width and Rs is significantly non-uniform. FIG. 1I shows an unfilled and filled narrow asymmetric trench structure typical of DRAM bWL. The unfilled features are adjacent and generally V-shaped, having sloped sidewalls. The features widen from the feature bottom to the feature top. After tungsten fill, severe line bending is observed. Without being bound by a particular theory, it is believed that a cohesive force between opposing surfaces of a trench pulls the trench sides together. This phenomena is illustrated in FIG. 1J, and may be characterized as “zipping up” the feature. As the feature is filled, more force is exerted from a center axis of the feature, causing line bending. FIG. 1K illustrates the interatomic force as a function of tungsten-tungsten bond radius, r . As can be seen, a cohesive force exists at certain values of r . In some embodiments, the pitch (feature to feature distance from feature center axes) is below 50 nm, below 40 nm, or below 30 nm.

FIGS. 2-4 provide overviews of various processes of tungsten feature fill that can address the above issues, with examples of tungsten fill of various features described with reference to FIGS. 5-7.

FIG. 2 is a process flow diagram illustrating certain operations in a method of filling a feature with tungsten. The method begins at a block **201** with selective inhibition of a feature. Selective inhibition, which may also be referred to as selective passivation, differential inhibition, or differential passivation, involves inhibiting subsequent tungsten nucleation on a portion of the feature, while not inhibiting nucleation (or inhibiting nucleation to a lesser extent) on the remainder of the feature. For example, in some embodiments, a feature is selectively inhibited at a feature opening, while nucleation inside the feature is not inhibited. Selective inhibition is described further below, and can involve, for example, selectively exposing a portion of the feature to activated species of a plasma. In certain embodiments, for example, a feature opening is selectively exposed to a plasma generated from molecular nitrogen gas. As discussed further below, a desired inhibition profile in a feature can be formed by appropriately selecting one or more of inhibition chemistry, substrate bias power, plasma power, process pressure, exposure time, and other process parameters.

Once the feature is selectively inhibited, the method can continue at block **203** with selective deposition of tungsten according to the inhibition profile. Block **203** may involve one or more chemical vapor deposition (CVD) and/or atomic layer deposition (ALD) processes, including thermal, plasma-enhanced CVD and/or ALD processes. The deposition is selective in that the tungsten preferentially grows on the lesser- and non-inhibited portions of the feature. In some embodiments, block **203** involves selectively depositing tungsten in a bottom or interior portion of the feature until a constriction is reached or passed.

After selective deposition according to the inhibition profile is performed, the method can continue at block **205** with filling the rest of the feature. In certain embodiments, block **205** involves a CVD process in which a tungsten-containing precursor is reduced by hydrogen to deposit tungsten. While tungsten hexafluoride (WF_6) is often used, the process may be performed with other tungsten precursors, including, but not limited to, tungsten hexachloride (WCl_6), organo-metallic precursors, and precursors that are free of fluorine such as MDNOW (methylcyclopentadienyl-dicarbonylnitrosyl-tungsten) and EDNOW (ethylcyclopentadienyl-dicarbonylnitrosyl-tungsten). In addition, while hydrogen can be used as the reducing agent in the CVD deposition, other reducing agents including silane may be used in addition or instead of hydrogen. In another embodiment, tungsten hexacarbonyl ($W(CO)_6$) may be used with or without a reducing agent. Unlike with ALD and pulsed nucleation layer (PNL) processes described further below, in a CVD technique, the WF_6 and H_2 or other reactants are simultaneously introduced into the reaction chamber. This produces a continuous chemical reaction of mix reactant gases that continuously forms tungsten film on the substrate surface. Methods of depositing tungsten films using CVD are described in U.S. patent application Ser. Nos. 12/202,126, 12/755,248 and 12/755,259, which are incorporated by reference herein in their entireties for the purposes of describing tungsten deposition processes. According to various embodiments, the methods described herein are not limited to a particular method of filling a feature but may include any appropriate deposition technique.

In some embodiments, block **205** may involve continuing a CVD deposition process started at block **203**. Such a CVD process may result in deposition on the inhibited portions of the feature, with nucleation occurring more slowly than on the non-inhibited portions of the feature. In some embodi-

ments, block **205** may involve deposition of a tungsten nucleation layer over at least the inhibited portions of the feature.

According to various embodiments, the feature surface that is selectively inhibited can be a barrier or liner layer, such as a metal nitride layer, or it can be a layer deposited to promote nucleation of tungsten. FIG. 3 shows an example of a method in which a tungsten nucleation layer is deposited in the feature prior to selective inhibition. The method begins at block **301** with deposition of the thin conformal layer of tungsten in the feature. The layer can facilitate subsequent deposition of bulk tungsten-containing material thereon. In certain embodiments, the nucleation layer is deposited using a PNL technique. In a PNL technique, pulses of a reducing agent, purge gases, and tungsten-containing precursor can be sequentially injected into and purged from the reaction chamber. The process is repeated in a cyclical fashion until the desired thickness is achieved. PNL broadly embodies any cyclical process of sequentially adding reactants for reaction on a semiconductor substrate, including ALD techniques. PNL techniques for depositing tungsten nucleation layers are described in U.S. Pat. Nos. 6,635,965; 7,589,017; 7,141,494; 7,772,114; 8,058,170 and in U.S. patent application Ser. Nos. 12/755,248 and 12/755,259, which are incorporated by reference herein in their entireties for the purposes of describing tungsten deposition processes. Block **301** is not limited to a particular method of tungsten nucleation layer deposition, but includes PNL, ALD, CVD, and physical vapor deposition (PVD) techniques for depositing a thin conformal layer. The nucleation layer can be sufficiently thick to fully cover the feature to support high quality bulk deposition; however, because the resistivity of the nucleation layer is higher than that of the bulk layer, the thickness of the nucleation layer may be minimized to keep the total resistance as low as possible. Example thicknesses of films deposited in block **301** can range from less than 10 Å to 100 Å. After deposition of the thin conformal layer of tungsten in block **301**, the method can continue with blocks **201**, **203**, and **205** as described above with reference to FIG. 2. An example of filling a feature according to a method of FIG. 3 is described below with reference to FIG. 5.

FIG. 4 shows an example of a method in which completing filling the feature (e.g., block **205** in FIG. 2 or 3) can involve repeating selective inhibition and deposition operations. The method can begin at block **201**, as described above with respect to FIG. 2, in which the feature is selectively inhibited, and continue at block **203** with selective deposition according to the inhibition profile. Blocks **201** and **203** are then repeated one or more times (block **401**) to complete feature fill. An example of filling a feature according to a method of FIG. 4 is described below with reference to FIG. 6.

Still further, selective inhibition can be used in conjunction with selective deposition. Selective deposition techniques are described in U.S. Provisional Patent Application No. 61/616,377, referenced above.

According to various embodiments, selective inhibition can involve exposure to activated species that passivate the feature surfaces. For example, in certain embodiments, a tungsten (W) surface can be passivated by exposure to a nitrogen-based or hydrogen-based plasma. In some embodiments, inhibition can involve a chemical reaction between activated species and the feature surface to form a thin layer of a compound material such as tungsten nitride (WN) or tungsten carbide (WC). In some embodiments, inhibition can involve a surface effect such as adsorption that passivates the surface without forming a layer of a compound

material. Activated species may be formed by any appropriate method including by plasma generation and/or exposure to ultraviolet (UV) radiation. In some embodiments, the substrate including the feature is exposed to a plasma generated from one or more gases fed into the chamber in which the substrate sits. In some embodiments, one or more gases may be fed into a remote plasma generator, with activated species formed in the remote plasma generator fed into a chamber in which the substrate sits. The plasma source can be any type of source including radio frequency (RF) plasma source or microwave source. The plasma can be inductively and/or capacitively-coupled. Activated species can include atomic species, radical species, and ionic species. In certain embodiments, exposure to a remotely-generated plasma includes exposure to radical and atomized species, with substantially no ionic species present in the plasma such that the inhibition process is not ion-mediated. In other embodiments, ion species may be present in a remotely-generated plasma. In certain embodiments, exposure to an in-situ plasma involves ion-mediated inhibition. For the purposes of this application, activated species are distinguished from recombined species and from the gases initially fed into a plasma generator.

Inhibition chemistries can be tailored to the surface that will be subsequently exposed to deposition gases. For tungsten (W) surfaces, as formed for example in a method described with reference to FIG. 3, exposure to nitrogen-based and/or hydrogen-based plasmas inhibits subsequent tungsten deposition on the W surfaces. Other chemistries that may be used for inhibition of tungsten surfaces include oxygen-based plasmas and hydrocarbon-based plasmas. For example, molecular oxygen or methane may be introduced to a plasma generator.

As used herein, a nitrogen-based plasma is a plasma in which the main non-inert component is nitrogen. An inert component such as argon, xenon, or krypton may be used as a carrier gas. In some embodiments, no other non-inert components are present in the gas from which the plasma is generated except in trace amounts. Similarly, a hydrogen-based plasma is a plasma in which the main non-inert component is hydrogen. In some embodiments, inhibition chemistries may be nitrogen-containing, hydrogen-containing, oxygen-containing, and/or carbon-containing, with one or more additional reactive species present in the plasma. For example, U.S. patent application Ser. No. 13/016,656, incorporated by reference herein, describes passivation of a tungsten surface by exposure to nitrogen trifluoride (NF₃). Similarly, fluorocarbons such as CF₄ or C₂F₈ may be used. However, in certain embodiments, the inhibition species are fluorine-free to prevent etching during selective inhibition.

In certain embodiments, UV radiation may be used in addition to or instead of plasma to provide activated species. Gases may be exposed to UV light upstream of and/or inside a reaction chamber in which the substrate sits. Moreover, in certain embodiments, non-plasma, non-UV, thermal inhibition processes may be used. In addition to tungsten surfaces, nucleation may be inhibited on liner/barrier layers surfaces such as TiN and/or WN surfaces. Any chemistry that passivates these surfaces may be used. For TiN and WN, this can include exposure to nitrogen-based or nitrogen-containing chemistries. In certain embodiments, the chemistries described above for W may also be employed for TiN, WN, or other liner layer surfaces.

Tuning an inhibition profile can involve appropriately controlling an inhibition chemistry, substrate bias power, plasma power, process pressure, exposure time, and other process parameters. For in situ plasma processes (or other

processes in which ionic species are present), a bias can be applied to the substrate. Substrate bias can, in some embodiments, significantly affect an inhibition profile, with increasing bias power resulting in active species deeper within the feature. For example, 100 W DC bias on a 300 mm substrate may result inhibition the top half of a 1500 nm deep structure, while a 700 W bias may result in inhibition of the entire structure. The absolute bias power appropriate a particular selective inhibition will depend on the substrate size, the system, plasma type, and other process parameters, as well as the desired inhibition profile, however, bias power can be used to tune top-to-bottom selectivity, with decreasing bias power resulting in higher selectivity. For 3-D structures in which selectivity is desired in a lateral direction (tungsten deposition preferred in the interior of the structure), but not in a vertical direction, increased bias power can be used to promote top-to-bottom deposition uniformity.

While bias power can be used in certain embodiments as the primary or only knob to tune an inhibition profile for ionic species, in certain situations, other performing selective inhibition uses other parameters in addition to or instead of bias power. These include remotely generated non-ionic plasma processes and non-plasma processes. Also, in many systems, a substrate bias can be easily applied to tune selectivity in vertical but not lateral direction. Accordingly, for 3-D structures in which lateral selectivity is desired, parameters other than bias may be controlled, as described above.

Inhibition chemistry can also be used to tune an inhibition profile, with different ratios of active inhibiting species used. For example, for inhibition of W surfaces, nitrogen may have a stronger inhibiting effect than hydrogen; adjusting the ratio of N₂ and H₂ gas in a forming gas-based plasma can be used to tune a profile. The plasma power may also be used to tune an inhibition profile, with different ratios of active species tuned by plasma power. Process pressure can be used to tune a profile, as pressure can cause more recombination (deactivating active species) as well as pushing active species further into a feature. Process time may also be used to tune inhibition profiles, with increasing treatment time causing inhibition deeper into a feature.

In some embodiments, selective inhibition can be achieved by performing operation 203 in a mass transport limited regime. In this regime, the inhibition rate inside the feature is limited by amounts of and/or relative compositions of different inhibition material components (e.g., an initial inhibition species, activated inhibition species, and recombined inhibition species) that diffuse into the feature. In certain examples, inhibition rates depend on various components' concentrations at different locations inside the feature.

Mass transport limiting conditions may be characterized, in part, by overall inhibition concentration variations. In certain embodiments, a concentration is less inside the feature than near its opening resulting in a higher inhibition rate near the opening than inside. This in turn leads to selective inhibition near the feature opening. Mass transport limiting process conditions may be achieved by supplying limited amounts of inhibition species into the processing chamber (e.g., use low inhibition gas flow rates relative to the cavity profile and dimensions), while maintaining relative high inhibition rates near the feature opening to consume some activated species as they diffuse into the feature. In certain embodiment, a concentration gradient is substantial, which may be caused relatively high inhibition kinetics and relatively low inhibition supply. In certain embodi-

ments, an inhibition rate near the opening may also be mass transport limited, though this condition is not required to achieve selective inhibition.

In addition to the overall inhibition concentration variations inside features, selective inhibition may be influenced by relative concentrations of different inhibition species throughout the feature. These relative concentrations in turn can depend on relative dynamics of dissociation and recombination processes of the inhibition species. As described above, an initial inhibition material, such as molecular nitrogen, can be passed through a remote plasma generator and/or subjected to an in-situ plasma to generate activated species (e.g., atomic nitrogen, nitrogen ions). However, activated species may recombine into less active recombined species (e.g., nitrogen molecules) and/or react with W, WN, TiN, or other feature surfaces along their diffusion paths. As such, different parts of the feature may be exposed to different concentrations of different inhibition materials, e.g., an initial inhibition gas, activated inhibition species, and recombined inhibition species. This provides additional opportunities for controlling selective inhibition. For example, activated species are generally more reactive than initial inhibition gases and recombined inhibition species. Furthermore, in some cases, the activated species may be less sensitive to temperature variations than the recombined species. Therefore, process conditions may be controlled in such a way that removal is predominantly attributed to activated species. As noted above, some species may be more reactive than others. Furthermore, specific process conditions may result in activated species being present at higher concentrations near features' openings than inside the features. For example, some activated species may be consumed (e.g., reacted with feature surface materials and/or adsorbed on the surface) and/or recombined while diffusing deeper into the features, especially in small high aspect ratio features. Recombination of activated species can also occur outside of features, e.g., in the showerhead or the processing chamber, and can depends on chamber pressure. Therefore, chamber pressure may be specifically controlled to adjust concentrations of activated species at various points of the chamber and features.

Flow rates of the inhibition gas can depend on a size of the chamber, reaction rates, and other parameters. A flow rate can be selected in such a way that more inhibition material is concentrated near the opening than inside the feature. In certain embodiments, these flow rates cause mass-transport limited selective inhibition. For example, a flow rate for a 195-liter chamber per station may be between about 25 sccm and 10,000 sccm or, in more specific embodiments, between about 50 sccm and 1,000 sccm. In certain embodiments, the flow rate is less than about 2,000 sccm, less than about 1,000 sccm, or more specifically less than about 500 sccm. It should be noted that these values are presented for one individual station configured for processing a 300-mm substrate. These flow rates can be scaled up or down depending on a substrate size, a number of stations in the apparatus (e.g., quadruple for a four station apparatus), a processing chamber volume, and other factors.

In certain embodiments, the substrate can be heated up or cooled down before selective inhibition. Various devices may be used to bring the substrate to the predetermined temperature, such as a heating or cooling element in a station (e.g., an electrical resistance heater in stalled in a pedestal or a heat transfer fluid circulated through a pedestal), infrared lamps above the substrate, igniting plasma, etc.

A predetermined temperature for the substrate can be selected to induce a chemical reaction between the feature

surface and inhibition species and/or promote adsorption of the inhibition species, as well as to control the rate of the reaction or adsorption. For example, a temperature may be selected to have high reaction rate such that more inhibition occurs near the opening than inside the feature. Furthermore, a temperature may be also selected to control recombination of activated species (e.g., recombination of atomic nitrogen into molecular nitrogen) and/or control which species (e.g., activated or recombined species) contribute predominantly to inhibition. In certain embodiments, a substrate is maintained at less than about 300° C., or more particularly at less than about 250° C., or less than about 150° C., or even less than about 100° C. In other embodiments, a substrate is heated to between about 300° C. and 450° C. or, in more specific embodiments, to between about 350° C. and 400° C. Other temperature ranges may be used for different types of inhibition chemistries. Exposure time can also be selected to cause selective inhibition. Example exposure times can range from about 10 s to 500 s, depending on desired selectivity and feature depth.

In some embodiments, thermal inhibition processes are provided. Thermal inhibition processes generally involve exposing the feature to a nitrogen-containing compound such as ammonia (NH₃) or hydrazine (N₂H₄) to non-conformally inhibit the feature near the feature opening. In some embodiments, the thermal inhibition processes are performed at temperatures ranging from 250° C. to 450° C. At these temperatures, exposure of a previously formed tungsten nucleation layer to NH₃ results in an inhibition effect. Other potentially inhibiting chemistries such as nitrogen (N₂) or hydrogen (H₂) may be used for thermal inhibition at higher temperatures (e.g., 900° C.). For many applications, however, these high temperatures exceed the thermal budget. In addition to ammonia, other hydrogen-containing nitriding agents such as hydrazine may be used at lower temperatures appropriate for back end of line (BEOL) applications.

As the thermal inhibition processes do not use a plasma, a bias power on the substrate cannot be used to tune the inhibition profile. However, by appropriately tuning one or more of chamber pressure, flow rate, dose time, and temperature, the inhibition profile can be tuned as desired. As described above, in some embodiments, a mass transport limited regime is employed. Chamber pressure may range from 0.5 Torr to 40 Torr in some embodiments. As noted above, flow rates of the inhibition gas can depend on a size of the chamber, reaction rates, and other parameters. A flow rate can be selected in such a way that more inhibition material is concentrated near the opening than inside the feature. In certain embodiments, these flow rates cause mass-transport limited selective inhibition.

Increased pressures and decreased flow rates and dose times result in more non-conformal (i.e., more selective to the feature opening) inhibition profile. Higher pressures result in lower mean free path, while lower flow rates and dose times limit the amount of molecules to be consumed. Increased temperatures result in a more non-conformal inhibition profile with more inhibition molecules consumed at the top of the feature. The profile may be tuned as described above depending on if and where a pinch point is in a feature. Example dose times range from 0.5 second to 10 seconds.

In some embodiments, inhibition can involve a chemical reaction between the thermal inhibitor species and the feature surface to form a thin layer of WN compound material. In some embodiments, inhibition can involve a

surface effect such as adsorption that passivates the surface without forming a layer of a compound material.

If a tungsten nucleation layer is present, it may be exposed to NH₃ or other inhibition vapor to selectively inhibit the feature at the feature top. In some embodiments, if a bulk tungsten or tungsten-containing layer is present, a reducing agent/tungsten-containing precursor/nitrogen-containing inhibition chemistry may be employed to form WN on the bulk layer. These reactants may be introduced in sequence (e.g., B₂H₆/WF₆/NH₃ pulses) or simultaneously. Any appropriate reducing agent (e.g., diborane or silane) and any appropriate tungsten-containing precursor (e.g., tungsten hexafluoride or tungsten hexacarbonyl) may be used. A thermal process may be used to avoid damage that may arise from the use of a plasma.

As described above, aspects of the invention can be used for VNAND wordline (WL) fill. While the below discussion provides a framework for various methods, the methods are not so limited and can be implemented in other applications as well, including logic and memory contact fill, DRAM buried wordline fill, vertically integrated memory gate/wordline fill, and 3D integration (TSV).

FIG. 1F, described above, provides an example of a VNAND wordline structure to be filled. As discussed above, feature fill of these structures can present several challenges including constrictions presented by pillar placement. In addition, a high feature density can cause a loading effect such that reactants are used up prior to complete fill.

Various methods are described below for void-free fill through the entire WL. In certain embodiments, low resistivity tungsten is deposited. FIG. 5 shows a sequence in which non-conformal selective inhibition is used to fill in the interior of the feature before pinch off. In FIG. 5, a structure 500 is provided with a liner layer surface 502. The liner layer surface 502 may be for example, TiN or WN. Next, a W nucleation layer 504 is conformally deposited on the liner layer 502. A PNL process as described above can be used. Note that in some embodiments, this operation of depositing a conformal nucleation layer may be omitted. Next, the structure is exposed to an inhibition chemistry to selectively inhibit portions 506 of the structure 500. In this example, the portions 508 through the pillar constrictions 151 are selectively inhibited. Inhibition can involve for example, exposure to a direct (in-situ) plasma generated from a gas such as N₂, H₂, forming gas, NH₃, O₂, CH₄, etc. Other methods of exposing the feature to inhibition species are described above. Next, a CVD process is performed to selectively deposit tungsten in accordance with the inhibition profile: bulk tungsten 510 is preferentially deposited on the non-inhibited portions of the nucleation layer 504, such that hard-to-fill regions behind constrictions are filled. The remainder of the feature is then filled with bulk tungsten 510. As described above with reference to FIG. 2, the same CVD process used to selectively deposit tungsten may be used to remainder of the feature, or a different CVD process using a different chemistry or process conditions and/or performed after a nucleation layer is deposited may be used.

In some embodiments, methods described herein may be used for tungsten via fill. FIG. 6 shows an example of a feature hole 105 including an under-layer 113, which can be, for example, a metal nitride or other barrier layer. A tungsten layer 653 is conformally deposited in the feature hole 10, for example, by a PNL and/or CVD method. (Note that while the tungsten layer 653 is conformally deposited in the feature hole 105 in the example of FIG. 6, in some other embodiments, tungsten nucleation on the under-layer 113 can be selectively inhibited prior to selective deposition of

13

the tungsten layer 653.) Further deposition on the tungsten layer 653 is then selectively inhibited, forming inhibited portion 655 of the tungsten layer 653 near the feature opening. Tungsten is then selectively deposited by a PNL and/or CVD method in accordance with the inhibition profile such that tungsten is preferentially deposited near the bottom and mid-section of the feature. Deposition continues, in some embodiments with one or more selective inhibition cycles, until the feature is filled. As described above, in some embodiments, the inhibition effect at the feature top can be overcome by a long enough deposition time, while in some embodiments, an additional nucleation layer deposition or other treatment may be performed to lessen or remove the passivation at the feature opening once deposition there is desired. Note that in some embodiments, feature fill may still include formation of a seam, such as seam 657 depicted in FIG. 6. In other embodiments, the feature fill may be void-free and seam-free. Even if a seam is present, it may be smaller than obtained with a conventionally filled feature, reducing the problem of coring during CMP. The sequence depicted in the example of FIG. 6 ends post-CMP with a relatively small void present.

In some embodiments, the processes described herein may be used advantageously even for features that do not have constrictions or possible pinch-off points. For example, the processes may be used for bottom-up, rather than conformal, fill of a feature. FIG. 7 depicts a sequence in which a feature 700 is filled by a method according to certain embodiments. A thin conformal layer of tungsten 753 is deposited initially, followed by selective inhibition to form inhibited portions 755, layer 753 at the bottom of the feature not treated. CVD deposition results in a bulk film 757 deposited on at the bottom of the feature. This is then followed by repeated cycles of selective CVD deposition and selective inhibition until the feature is filled with bulk tungsten 757. Because nucleation on the sidewalls of the feature is inhibited except near the bottom of the feature, fill is bottom-up. In some embodiments, different parameters may be used in successive inhibitions to tune the inhibition profile appropriately as the bottom of the feature grows closer to the feature opening. For example, a bias power and/or treatment time may be decreased in successive inhibition treatments.

In some embodiments, the above-described processes may be performed on multiple adjacent bWL trenches without line-bending. Example opening widths range from 14-19 nm, with center-to-center spacing being 40-50 nm. Widths (e.g., 10-30 nm) and spacings (e.g., 30-60 nm) outside these ranges may be used.

FIG. 8 shows an example of trenches filled according to a method as shown in FIG. 3. A thin conformal layer is deposited in the trenches as described with respect to block 301. (The thin conformal layer is shown as dotted for ease of visualization and not to suggest that it is necessarily discontinuous. It may be a continuous thin conformal layer.) The trenches are then selectively inhibited as described in block 201, with the solid line overlying the dotted line indicated the inhibited portion of the thin conformal layer. The selective inhibition is followed by selective deposition in accordance with the inhibition profile as described with respect to block 203. The remaining part of the trenches are then filled as described with respect to block 205. In some embodiments, the inhibition is performed using a remote plasma or thermal inhibition to prevent line bending due to plasma charging.

The reduction in line bending due to the selective inhibition is described with respect to FIGS. 1K and 9. In FIG.

14

9, the second deposition step, in which tungsten is deposited in accordance with the inhibition profile, is shown. The inhibited sidewall surfaces (also referred to as opposing surface) do not grow toward each other (or grow more slowly than they otherwise would.) Referring to FIG. 1K, the large separation means a low force as indicated by the arrow.

As described above, the deposition operations in FIG. 8 may be CVD or ALD. In some embodiments, one or more deposition processes as described in U.S. patent application Ser. No. 14/723,270, incorporated by reference herein, may be used.

EXPERIMENTAL

3D VNAND features similar to the schematic depiction in FIG. 1F were exposed to plasmas generated from N_2H_2 gas after deposition of an initial tungsten seed layer. The substrate was biased with a DC bias, with bias power varied from 100 W to 700 W and exposure time varied between 20 s and 200 s. Longer time resulted in deeper and wider inhibition, with higher bias power resulting in deeper inhibition.

Table 1 shows effect of treatment time. All inhibition treatments used exposure to a direct LFRF 2000 W N_2H_2 plasma with a DC bias of 100 W on the substrate.

TABLE 1

Effect of treatment time on inhibition profile				
	Initial Tungsten Layer	Inhibition Treatment Time	Subsequent Deposition	Selective Deposition
A	Nucleation + 30 s CVD at 300° C.	None	400 s CVD at 300° C.	Non-selective deposition
B	same as A	60 s	same as A	Non-selective deposition
C	same as A	90 s	same as A	Yes - deposition only from bottom of feature to slightly less than vertical midpoint. Lateral deposition (wider) at bottom of feature.
D	same as A	140 s	same as A	No deposition

While varying treatment time resulted in vertical and lateral tuning of inhibition profile as described in Table 1 (split C), varying bias power correlated higher to vertical tuning of inhibition profile, with lateral variation a secondary effect.

As described above, the inhibition effect may be overcome by certain CVD conditions, including longer CVD time and/or higher temperatures, more aggressive chemistry, etc. Table 2 below, shows the effect of CVD time on selective deposition.

15

TABLE 2

Effect of CVD time on selective deposition				
	Initial Tungsten Layer	Inhibition Treatment	Subsequent CVD Deposition Time (300° C.)	Selective Deposition
E	Nucleation + 30 s CVD at 300° C.	H ₂ N ₂ 2000 W RF direct plasma, 90 s, 100 W DC bias	0	no deposition
F	same as E	same as E	200 s	Yes - small amount of deposition extending about 1/6 height of feature from bottom
G	same as E	same as E	400 s	Yes - deposition only from bottom of feature to slightly less than vertical midpoint. Lateral deposition wider at bottom of feature.
H	same as E	same as E	700 s	Yes - deposition through full height of feature, with lateral deposition wider at bottom of feature

FIG. 10A shows images of fill improvement from a thermal (no plasma) inhibition process. Features having openings of approximately 150 nm and over 30:1 aspect ratios were filled with tungsten, with no inhibition (1010) and thermal inhibition using NH₃ (1020). FIG. 10B shows side-by-side enlarged comparisons of feature fill with no inhibition (1110) and thermal inhibition (1120) at the top, middle, and bottom of two features from the images in FIG. 10A.

Line-to-line non-uniformity was measured for line trenches filled using no inhibition as compared to depositions using remote plasma inhibition. FIG. 11 shows that increasing remote plasma inhibition according to a method as discussed with respect to FIGS. 2 and 8 results in decreasing non-uniformity.

Apparatus

Any suitable chamber may be used to implement this novel method. Examples of deposition apparatuses include various systems, e.g., ALTUS and ALTUS Max, available from Novellus Systems, Inc. of San Jose, California, or any of a variety of other commercially available processing systems.

FIG. 12 illustrates a schematic representation of an apparatus 1200 for processing a partially fabricated semiconductor substrate in accordance with certain embodiments. The apparatus 1200 includes a chamber 1218 with a pedestal 1220, a shower head 1214, and an in-situ plasma generator 1216. The apparatus 1200 also includes a system controller 1222 to receive input and/or supply control signals to various devices.

In certain embodiments, an inhibition gas and, if present, inert gases, such as argon, helium and others, can be supplied to the remote plasma generator 1206 from a source 1202, which may be a storage tank. Any suitable remote plasma generator may be used for activating the etchant before introducing it into the chamber 1218. For example, a Remote Plasma Cleaning (RPC) units, such as ASTRON® i Type AX7670, ASTRON® e Type AX7680, ASTRON® ex Type AX7685, ASTRON® hf-s Type AX7645, all available

16

from MKS Instruments of Andover, Massachusetts, may be used. An RPC unit is typically a self-contained device generating weakly ionized plasma using the supplied etchant. Embedded into the RPC unit a high power RF generator provides energy to the electrons in the plasma. This energy is then transferred to the neutral inhibition gas molecules leading to temperature in the order of 2000K causing thermal dissociation of these molecules. An RPC unit may dissociate more than 60% of incoming molecules because of its high RF energy and special channel geometry causing the gas to adsorb most of this energy.

In certain embodiments, an inhibition gas is flown from the remote plasma generator 1206 through a connecting line 1208 into the chamber 1218, where the mixture is distributed through the shower head 1214. In other embodiments, an inhibition gas is flown into the chamber 1218 directly completely bypassing the remote plasma generator 1206 (e.g., the system 1200 does not include such generator). Alternatively, the remote plasma generator 1206 may be turned off while flowing the inhibition gas into the chamber 1218, for example, because activation of the inhibition gas is not needed or will be supplied by an in situ plasma generator. Inert gases 1212 may be mixed in some embodiments in a mixing bowl 1110.

The shower head 1214 or the pedestal 1220 typically may have an internal plasma generator 1216 attached to it. In one example, the generator 1216 is a High Frequency (HF) generator capable of providing between about 0 W and 10,000 W at frequencies between about 1 MHz and 100 MHz. In another example, the generator 1216 is a Low Frequency (LF) generator capable of providing between about 0 W and 10,000 W at frequencies as low as about 100 KHz. In a more specific embodiment, a HF generator may deliver between about 0 W to 5,000 W at about 13.56 MHz. The RF generator 1216 may generate in-situ plasma to active inhibition species. In certain embodiments, the RF generator 1216 can be used with the remote plasma generator 1206 or not used. In certain embodiments, no plasma generator is used during deposition.

The chamber 1218 may include a sensor 1224 for sensing various process parameters, such as degree of deposition, concentrations, pressure, temperature, and others. The sensor 1224 may provide information on chamber conditions during the process to the system controller 1222. Examples of the sensor 1224 include mass flow controllers, pressure sensors, thermocouples, and others. The sensor 1224 may also include an infra-red detector or optical detector to monitor presence of gases in the chamber and control measures.

Deposition and selective inhibition operations can generate various volatile species that are evacuated from the chamber 1218. Moreover, processing is performed at certain predetermined pressure levels the chamber 1218. Both of these functions are achieved using a vacuum outlet 1226, which may be a vacuum pump.

In certain embodiments, a system controller 1222 is employed to control process parameters. The system controller 1222 typically includes one or more memory devices and one or more processors. The processor may include a CPU or computer, analog and/or digital input/output connections, stepper motor controller boards, etc. Typically there will be a user interface associated with system controller 1222. The user interface may include a display screen, graphical software displays of the apparatus and/or process conditions, and user input devices such as pointing devices, keyboards, touch screens, microphones, etc.

In certain embodiments, the system controller **1222** controls the substrate temperature, inhibition gas flow rate, power output of the remote plasma generator **1206** and/or in situ plasma generator **1216**, pressure inside the chamber **1218** and other process parameters. The system controller **1222** executes system control software including sets of instructions for controlling the timing, mixture of gases, chamber pressure, chamber temperature, and other parameters of a particular process. Other computer programs stored on memory devices associated with the controller may be employed in some embodiments.

The computer program code for controlling the processes in a process sequence can be written in any conventional computer readable programming language: for example, assembly language, C, C++, Pascal, Fortran or others. Compiled object code or script is executed by the processor to perform the tasks identified in the program. The system software may be designed or configured in many different ways. For example, various chamber component subroutines or control objects may be written to control operation of the chamber components necessary to carry out the described processes. Examples of programs or sections of programs for this purpose include process gas control code, pressure control code, and plasma control code.

The controller parameters relate to process conditions such as, for example, timing of each operation, pressure inside the chamber, substrate temperature, inhibition gas flow rates, etc. These parameters are provided to the user in the form of a recipe, and may be entered utilizing the user interface. Signals for monitoring the process may be provided by analog and/or digital input connections of the system controller **1222**. The signals for controlling the process are output on the analog and digital output connections of the apparatus **1200**.

Multi-Station Apparatus

FIG. **13A** shows an example of a multi-station apparatus **1300**. The apparatus **1300** includes a process chamber **1301** and one or more cassettes **1303** (e.g., Front Opening Unified Pods) for holding substrates to be processed and substrates that have completed processing. The chamber **1301** may have a number of stations, for example, two stations, three stations, four stations, five stations, six stations, seven stations, eight stations, ten stations, or any other number of stations. The number of stations is usually determined by a complexity of the processing operations and a number of these operations that can be performed in a shared environment. FIG. **13A** illustrates the process chamber **1301** that includes six stations, labeled **1311** through **1316**. All stations in the multi-station apparatus **1300** with a single process chamber **1303** are exposed to the same pressure environment. However, each station may have a designated reactant distribution system and local plasma and heating conditions achieved by a dedicated plasma generator and pedestal, such as the ones illustrated in FIG. **13**.

A substrate to be processed is loaded from one of the cassettes **1303** through a load-lock **1305** into the station **1311**. An external robot **1307** may be used to transfer the substrate from the cassette **1303** and into the load-lock **1305**. In the depicted embodiment, there are two separate load locks **1305**. These are typically equipped with substrate transferring devices to move substrates from the load-lock **1305** (once the pressure is equilibrated to a level corresponding to the internal environment of the process chamber **1303**) into the station **1311** and from the station **1316** back into the load-lock **1305** for removal from the processing chamber **1303**. A mechanism **1309** is used to transfer sub-

strates among the processing stations **1311-1316** and support some of the substrates during the process as described below.

In certain embodiments, one or more stations may be reserved for heating the substrate. Such stations may have a heating lamp (not shown) positioned above the substrate and/or a heating pedestal supporting the substrate similar to one illustrated in FIG. **12**. For example, a station **1311** may receive a substrate from a load-lock and be used to pre-heat the substrate before being further processed. Other stations may be used for filling high aspect ratio features including deposition and selective inhibition operations.

After the substrate is heated or otherwise processed at the station **1311**, the substrate is moved successively to the processing stations **1312**, **1313**, **1314**, **1315**, and **1316**, which may or may not be arranged sequentially. The multi-station apparatus **1300** can be configured such that all stations are exposed to the same pressure environment. In so doing, the substrates are transferred from the station **1311** to other stations in the chamber **1301** without a need for transfer ports, such as load-locks.

In certain embodiments, one or more stations may be used to fill features with tungsten-containing materials. For example, stations **1312** may be used for an initial deposition operation, station **1313** may be used for a corresponding selective inhibition operation. In the embodiments where a deposition-inhibition cycle is repeated, stations **1314** may be used for another deposition operations and station **1315** may be used for another inhibition operation. Section **1316** may be used for the final filling operation. It should be understood that any configurations of station designations to specific processes (heating, filling, and removal) may be used. In some implementations, any of the stations can be dedicated to one or more of PNL (or ALD) deposition, selective inhibition, and CVD deposition.

As an alternative to the multi-station apparatus described above, the method may be implemented in a single substrate chamber or a multi-station chamber processing a substrate(s) in a single processing station in batch mode (i.e., non-sequential). In this aspect of the invention, the substrate is loaded into the chamber and positioned on the pedestal of the single processing station (whether it is an apparatus having only one processing station or an apparatus having multi-stations running in batch mode). The substrate may be then heated and the deposition operation may be conducted. The process conditions in the chamber may be then adjusted and the selective inhibition of the deposited layer is then performed. The process may continue with one or more deposition-inhibition cycles (if performed) and with the final filling operation all performed on the same station. Alternatively, a single station apparatus may be first used to perform only one of the operation in the new method (e.g., depositing, selective inhibition, final filling) on multiple substrates after which the substrates may be returned back to the same station or moved to a different station (e.g., of a different apparatus) to perform one or more of the remaining operations.

Multi-Chamber Apparatus

FIG. **13B** is a schematic illustration of a multi-chamber apparatus **1320** that may be used in accordance with certain embodiments. As shown, the apparatus **1320** has three separate chambers **1321**, **1323**, and **1325**. Each of these chambers is illustrated with two pedestals. It should be understood that an apparatus may have any number of chambers (e.g., one, two, three, four, five, six, etc.) and each chamber may have any number of chambers (e.g., one, two, three, four, five, six, etc.). Each chamber **1321-1325** has its

own pressure environment, which is not shared between chambers. Each chamber may have one or more corresponding transfer ports (e.g., load-locks). The apparatus may also have a shared substrate handling robot 1327 for transferring substrates between the transfer ports one or more cassettes 1329.

As noted above, separate chambers may be used for depositing tungsten containing materials and selective inhibition of these deposited materials in later operations. Separating these two operations into different chambers can help to substantially improve processing speeds by maintaining the same environmental conditions in each chamber. A chamber does not need to change its environment from conditions used for deposition to conditions used for selective inhibition and back, which may involve different chemistries, different temperatures, pressures, and other process parameters. In certain embodiments, it is faster to transfer partially manufactured semiconductor substrates between two or more different chambers than changing environmental conditions of these chambers. Still further, one or more chambers may be used to etch. In some implementations, a controller is part of a system, which may be part of the above-described examples. Such systems can comprise semiconductor processing equipment, including a processing tool or tools, chamber or chambers, a platform or platforms for processing, and/or specific processing components (a wafer pedestal, a gas flow system, etc.). These systems may be integrated with electronics for controlling their operation before, during, and after processing of a semiconductor wafer or substrate. The electronics may be referred to as the "controller," which may control various components or subparts of the system or systems. The controller, depending on the processing requirements and/or the type of system, may be programmed to control any of the processes disclosed herein, including power, intensity, and exposure times. In an integrated tool, the controller may also control processes such as processing gases, temperature settings (e.g., heating and/or cooling), pressure settings, vacuum settings, power settings, radio frequency (RF) generator settings, RF matching circuit settings, frequency settings, flow rate settings, fluid delivery settings, positional and operation settings, wafer transfers into and out of a tool and other transfer tools and/or load locks connected to or interfaced with a specific system.

Broadly speaking, the controller may be defined as electronics having various integrated circuits, logic, memory, and/or software that receive instructions, issue instructions, control operation, enable cleaning operations, enable end-point measurements, and the like. The integrated circuits may include chips in the form of firmware that store program instructions, digital signal processors (DSPs), chips defined as application specific integrated circuits (ASICs), and/or one or more microprocessors, or microcontrollers that execute program instructions (e.g., software). The system software may be designed or configured in many different ways. For example, various chamber component subroutines or control objects may be written to control operation of the chamber components necessary to carry out the inventive processes. Examples of programs or sections of programs for this purpose include substrate positioning code, treatment compound control code, pressure control code, heater control code, and RF control code. In one embodiment, the controller includes instructions for performing processes of the disclosed embodiments according to methods described above. The computer program code for controlling the processes can be written in any conventional computer readable programming language: for example, assembly

language, C, C++, Pascal, Fortran, or others. Compiled object code or script is executed by the processor to perform the tasks identified in the program.

Program instructions may be instructions communicated to the controller in the form of various individual settings (or program files), defining operational parameters for carrying out a particular process on or for a semiconductor wafer or to a system. The operational parameters may, in some embodiments, be part of a recipe defined by process engineers to accomplish one or more processing steps during the fabrication of one or more layers, materials, metals, oxides, silicon, silicon dioxide, surfaces, circuits, and/or dies of a wafer.

The controller, in some implementations, may be a part of or coupled to a computer that is integrated with, coupled to the system, otherwise networked to the system, or a combination thereof. For example, the controller may be in the "cloud" or all or a part of a fab host computer system, which can allow for remote access of the wafer processing. The computer may enable remote access to the system to monitor current progress of fabrication operations, examine a history of past fabrication operations, examine trends or performance metrics from a plurality of fabrication operations, to change parameters of current processing, to set processing steps to follow a current processing, or to start a new process. In some examples, a remote computer (e.g. a server) can provide process recipes to a system over a network, which may include a local network or the Internet. The remote computer may include a user interface that enables entry or programming of parameters and/or settings, which are then communicated to the system from the remote computer. In some examples, the controller receives instructions in the form of data, which specify parameters for each of the processing steps to be performed during one or more operations. It should be understood that the parameters may be specific to the type of process to be performed and the type of tool that the controller is configured to interface with or control. Thus as described above, the controller may be distributed, such as by comprising one or more discrete controllers that are networked together and working towards a common purpose, such as the processes and controls described herein. An example of a distributed controller for such purposes would be one or more integrated circuits on a chamber in communication with one or more integrated circuits located remotely (such as at the platform level or as part of a remote computer) that combine to control a process on the chamber.

Without limitation, example systems may include a plasma etch chamber or module, a deposition chamber or module, a spin-rinse chamber or module, a metal plating chamber or module, a clean chamber or module, a bevel edge etch chamber or module, a physical vapor deposition (PVD) chamber or module, a chemical vapor deposition (CVD) chamber or module, an atomic layer deposition (ALD) chamber or module, an atomic layer etch (ALE) chamber or module, an ion implantation chamber or module, a track chamber or module, and any other semiconductor processing systems that may be associated or used in the fabrication and/or manufacturing of semiconductor wafers.

As noted above, depending on the process step or steps to be performed by the tool, the controller might communicate with one or more of other tool circuits or modules, other tool components, cluster tools, other tool interfaces, adjacent tools, neighboring tools, tools located throughout a factory, a main computer, another controller, or tools used in material transport that bring containers of wafers to and from tool locations and/or load ports in a semiconductor manufactur-

21

ing factory. There may be a user interface associated with controller. The user interface may include a display screen, graphical software displays of the apparatus and/or process conditions, and user input devices such as pointing devices, keyboards, touch screens, microphones, etc.

Patterning Method/Apparatus:

The apparatus/process described hereinabove may be used in conjunction with lithographic patterning tools or processes, for example, for the fabrication or manufacture of semiconductor devices, displays, LEDs, photovoltaic panels and the like. Typically, though not necessarily, such tools/processes will be used or conducted together in a common fabrication facility. Lithographic patterning of a film typically comprises some or all of the following steps, each step enabled with a number of possible tools: (1) application of photoresist on a workpiece, i.e., substrate, using a spin-on or spray-on tool; (2) curing of photoresist using a hot plate or furnace or UV curing tool; (3) exposing the photoresist to visible or UV or x-ray light with a tool such as a wafer stepper; (4) developing the resist so as to selectively remove resist and thereby pattern it using a tool such as a wet bench; (5) transferring the resist pattern into an underlying film or workpiece by using a dry or plasma-assisted etching tool; and (6) removing the resist using a tool such as an RF or microwave plasma resist stripper.

The invention claimed is:

1. A method comprising:

providing a substrate including a feature having a feature opening and a feature interior, the feature interior defined at least in part by a feature sidewall extending from the feature opening, and

exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process such that the feature sidewall is more inhibited near the feature opening than deeper into the feature.

2. The method of claim 1, wherein exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process comprises exposing the feature interior to a nitrogen-containing compound.

3. The method of claim 1, wherein exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process comprises exposing the feature interior to pulses of the gas-phase inhibition chemistry.

4. The method of claim 1, wherein exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process comprises exposing the feature interior to a nitrogen-containing and hydrogen-containing compound.

5. The method of claim 1, wherein exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process comprises passivating a surface without forming a compound material.

6. The method of claim 1, wherein exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process comprises passivating a surface by forming a compound material on the surface.

22

7. The method of claim 1, wherein exposing the feature interior to a gas-phase inhibition chemistry in a non-plasma process comprises exposing the feature interior to ammonia (NH_3).

8. The method of claim 1, further comprising selectively depositing metal in the feature such that more metal is deposited deeper in the feature interior than at the feature opening.

9. The method of claim 8, wherein the non-plasma process and selective deposition operations are performed in the same chamber.

10. The method of claim 8, wherein the non-plasma process and selective deposition operations are performed in different chambers.

11. The method of claim 1, wherein exposing the substrate to an inhibition chemistry comprises passivating a surface by forming a compound material on the surface.

12. A method comprising:

providing a substrate including a structure having a horizontally-oriented feature to be filled with a bulk material, the horizontally-oriented feature having a plurality of feature openings and an interior region accessible through the plurality of feature openings; and

exposing the substrate to an inhibition chemistry to inhibit growth of the bulk material in the horizontally-oriented feature such that growth is inhibited more at the plurality of feature openings than further in the interior region.

13. The method of claim 12, wherein the structure comprises a partially fabricated three-dimensional (3-D) NAND structure and the horizontally-oriented feature is one of a plurality of horizontally-oriented wordline features.

14. The method of claim 12, wherein exposing the substrate to an inhibition chemistry comprises exposing the substrate to a nitrogen-containing compound or nitrogen species.

15. The method of claim 12, further comprising selectively depositing metal in the interior region of the feature in accordance with the differential inhibition profile.

16. The method of claim 15, wherein exposure to the inhibition chemistry and selective deposition operations are performed in the same chamber.

17. The method of claim 15, wherein exposure to the inhibition chemistry and selective deposition operations are performed in different chambers.

18. The method of claim 12, exposing the substrate to an inhibition chemistry comprises passivating a surface without forming a compound material.

19. The method of claim 1, wherein the feature includes a second feature opening, wherein the feature sidewall extends from the feature opening to the second feature opening.

* * * * *